Communicative Efficiency in Adjective Ordering: Between Hierarchical Composition and Incremental Pragmatics

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Abstract

Adjective ordering preferences, such as placing size before colour, have been explained by two accounts of communicative efficiency. Hierarchical composition favours placing less subjective adjectives closer to the noun, where they restrict the interpretation of more subjective ones in the complete utterance; incremental pragmatics favours placing more discriminating adjectives earlier, where each adjective helps the listener narrow down the referent sooner. However, the two accounts can favour competing orders when the context makes the less subjective adjective the more discriminating one. Do speakers resolve this competition by evaluating the complete utterance, or each adjective as they produce it? We develop Rational Speech Act production models that formalise both accounts and span this range, from a global to an incremental speaker. Between these endpoints, a plan-guided speaker chooses each adjective for its immediate usefulness while looking ahead to the complete utterance, so that both pressures can influence adjective order. We found that the plan-guided speaker best predicted German production data, whereas modelling hierarchical composition added no predictive benefit. In simulations of communication, size-first order nevertheless helped the listener identify the referent once speakers could choose among all the utterances they produced, under every degree of planning and whether or not the model included hierarchical composition. Together, these findings motivate an account of boundedly rational communication in which partial planning supports good-enough production.

Keywords: adjective ordering, communicative efficiency, reference production, Rational Speech Act, Bayesian cognitive modelling

1 Introduction

Adjective ordering provides a window onto how stable order preferences interact with what is informative in a particular context. In languages with prenominal modification, robust ordering preferences favour size-first sequences such as *the big blue sticker* over *the blue big sticker* (Sproat and Shih, 1991; Trotzke and Wittenberg, 2019; Scontras, 2023). Communicative efficiency has been proposed as an explanation for both stable adjective-order preferences and their sensitivity to referential context (Scontras, Degen, and Goodman, 2017; Franke, Scontras, and Simonic, 2019; Hahn et al., 2018; Futrell, Dyer, and Scontras, 2020).

But what counts as efficient—and therefore early—in a complex referential setting? Two accounts give different answers, one based on hierarchical composition and one on incremental pragmatics. The first account, hierarchical composition, starts from the observation that more subjective adjectives tend to occur farther from the noun, which yields a size-first preference for size and colour (Scontras, Degen, and Goodman, 2017; Trainin and Shetreet, 2021). On this account, the noun-adjacent adjective restricts the context in which an outer adjective is interpreted (Franke, Scontras, and Simonic, 2019; Scontras, Degen, and Goodman, 2019; Scontras, Bar-Sever, et al., 2020). In *the big blue sticker*, *blue* combines with *sticker* before *big*, potentially narrowing the comparison class against which size is evaluated. Subjective meanings allow greater divergence between speakers' and listeners' interpretations, which can make reference less successful. Colour is less subjective than size, so evaluating *big* among the blue objects can reduce the risk that the listener misidentifies the target.

The second account, incremental pragmatics, concerns the order in which the listener receives information. Listeners use partial linguistic input and visual context to restrict candidate referents before a noun phrase is complete (Tanenhaus et al., 1995; Sedivy et al., 1999). Placing the more discriminating property first can facilitate this process, favouring colour-first order when colour identifies the target sooner (Fukumura, 2018; Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021).

The two accounts can therefore favour competing orders when the less subjective property is also the more discriminating one. When we apply both accounts to a context where colour alone identifies the target,[1] composition

[1] We call such contexts *colour-sufficient*. Together with size-sufficient contexts and contexts in which both properties are necessary, they

can still favour *the big blue sticker*, whereas incremental pragmatics favours *the blue big sticker*. The compositional benefit of size-first order depends on the complete phrase, whereas the incremental benefit of colour-first order depends on each adjective as it is produced. Scontras, Degen, and Goodman (2019, p. 7:19) noted that composition from the noun outwards stands at odds with the incremental uptake of words during reference resolution. Evidence for hierarchical composition comes mainly from ordering preferences in ratings and corpora (Scontras, Degen, and Goodman, 2017; Trainin and Shetreet, 2023; Scontras, 2023). Evidence for incremental pragmatics comes mainly from referential production and listeners' eye movements in visual displays (Fukumura, 2018; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). The two accounts have therefore rarely been compared within one model on the same data. Here, we compare the two accounts in reference production, where speakers formulate the utterance themselves.

In production, the competition between size-first and colour-first order can arise only when a speaker selects both size and colour, and not size or colour alone. In a colour-sufficient context, *blue* in *the blue sticker* may complete the utterance alone, so that size and colour need not be ordered at all. Speakers may nevertheless add *big*, even though it is redundant. Redundancy also occurs in size-sufficient contexts, where speakers may produce *the big blue sticker* although size alone identifies the target. Redundant colour, however, is far more common than redundant size, an asymmetry that has been attributed to the higher salience and lower context-dependence of colour (Tarenskeen, Broersma, and Geurts, 2015; Rubio-Fernández, 2016; Degen et al., 2020; Tourtouri, Delogu, and Crocker, 2021). Utterances with both adjectives may therefore arise mostly in size-sufficient contexts, and less often in the colour-sufficient contexts where the two accounts favour competing orders. Adding colour can help the listener, especially when size differences are hard to perceive (Rubio-Fernández, 2016; Degen et al., 2020). This observation motivates a second factor in our experiment, size discriminability, which varies how clearly the target stands out as big relative to its comparison class (Section 2). Size discriminability also separates the two accounts in contexts where both properties are needed. Low size discriminability makes it less certain whether the target counts as big, which favours size-first order under hierarchical composition, whereas high size discriminability makes size the more discriminating property, which favours size-first order under incremental pragmatics.

When speakers do produce both adjectives, previous studies show that the conventional size-first preference is so robust that, even when size is the least discriminating property, it is still preferred first, and reversed orders such as *the blue big sticker* are rare (Danks and Schwenk, 1972; Pechmann, 1989; Schlotterbeck and Wang, 2023). How can speakers keep such a stable order while adapting the content of an utterance to the context? The answer may depend on how far ahead an utterance is evaluated during production. A speaker may evaluate the complete utterance, or each adjective as it is produced. Again, in a colour-sufficient context, a speaker who has planned the complete utterance can begin with *big*, even though *big* is not the most salient adjective at that point, whereas a speaker who evaluates one adjective at a time is likely to begin with *blue* and stop there, because *blue* already identifies the target. This scope of evaluation also determines which efficiency benefit can shape the utterance. What a speaker produces, in both content and order, may thus depend on how much of the utterance is planned in advance.

At one end of this range, a global speaker compares complete utterances as wholes. At the other, a fully incremental speaker evaluates each next adjective against the alternatives available after the words already produced (Cohn-Gordon, Goodman, and Potts, 2019; Waldon and Degen, 2021). Research on bounded planning suggests that human speakers lie between these endpoints, beginning to speak before every part of an utterance has been prepared and continuing to plan during articulation (Ferreira and Swets, 2002; Konopka and Meyer, 2014). A speaker between these endpoints may choose each adjective for its immediate usefulness while looking ahead to the complete utterance, so that both pressures can influence adjective order. We call this intermediate hypothesis *plan-guided production*. A free parameter determines the degree of planning between these endpoints. Planning an utterance involves selecting properties as well as ordering them (Fukumura, Van Gompel, and Pickering, 2010; Van Gompel et al., 2019). The relevant production outcome is therefore the full distribution of one-, two-, and three-adjective utterances.

The Rational Speech Act (RSA) framework can predict this distribution by relating speaker choice to an utterance's informativeness and its production cost (Frank and Goodman, 2012; Goodman and Frank, 2016; Degen, 2023). Both efficiency accounts concern how informative an utterance is, so we formalise them within the same RSA model and compare them on the same production data. A joint formalisation can show whether the two pressures shape the same or different parts of an utterance, such as which adjectives are mentioned and how they are ordered. Schlotterbeck and Wang (2023) proposed an incremental RSA model that combined the two efficiency accounts. Their model compared two fixed adjective orders, which participants rated on a slider. Such ratings present both orders for comparison, so they cannot show which properties speakers would mention. It therefore remained open how the two accounts apply when speakers produce the utterances themselves. As the model combined both accounts, it also left open whether each account improves prediction on its own.

form the three levels of one experimental factor, referential context (Section 2).

Our central research question is: *How do speakers reconcile the pressures from hierarchical semantic composition and incremental pragmatics when selecting and ordering adjectives?* We address this question with two matched model comparisons, which we cross in a factorial design while holding all other components constant. The first targets the scope of evaluation that separates the two accounts. It asks which degree of planning best predicts the utterances speakers produce. The second asks whether narrowing the comparison class for size, as proposed by the hierarchical compositional account, improves prediction. We test these models on a German production experiment, in which speakers described a target object in a visual display. Their utterances show which content, length, and order speakers produce when they formulate an utterance themselves. Methodologically, we extend RSA production models from two fixed orders to the full set of utterances with one, two, or three adjectives for size, colour, and form. We estimate the models hierarchically across participants and compare them by how well they predict held-out data (Vehtari, Gelman, and Gabry, 2017).

On the compositional account, even a small advantage for identifying the referent could make one order conventional over repeated language use (Scontras, Degen, and Goodman, 2019, p. 7:19). Our experiment, by contrast, covers a limited set of displays, so predictive evidence from these displays need not reflect the advantage of an order across the many contexts of language use. Simulations that compared two complete orders found such an advantage for subjectivity-based order (Franke, Scontras, and Simonic, 2019). However, these simulations approximated communication by presupposing a speaker and a listener. We therefore implement both, by adding a pragmatic listener on top of the production models, and simulate communication across randomly generated contexts, to examine under which conditions hierarchical composition can favour size-first order when the speaker can also choose other utterances. The simulation shows that hierarchical composition shifts the listener's preference slightly towards size-first order, and that a larger size-first preference arises from the alternatives that speakers produce.

Figure 1 provides an overview of the study, including the approach, the models compared, and the main findings. Section 2 presents the production experiment, followed by the models and their degrees of planning in Section 3. Section 4 presents the predictive model comparisons; Section 5 examines the conditions under which communication favours size-first order. Section 6 discusses the implications for the two efficiency accounts and for bounded planning in production.

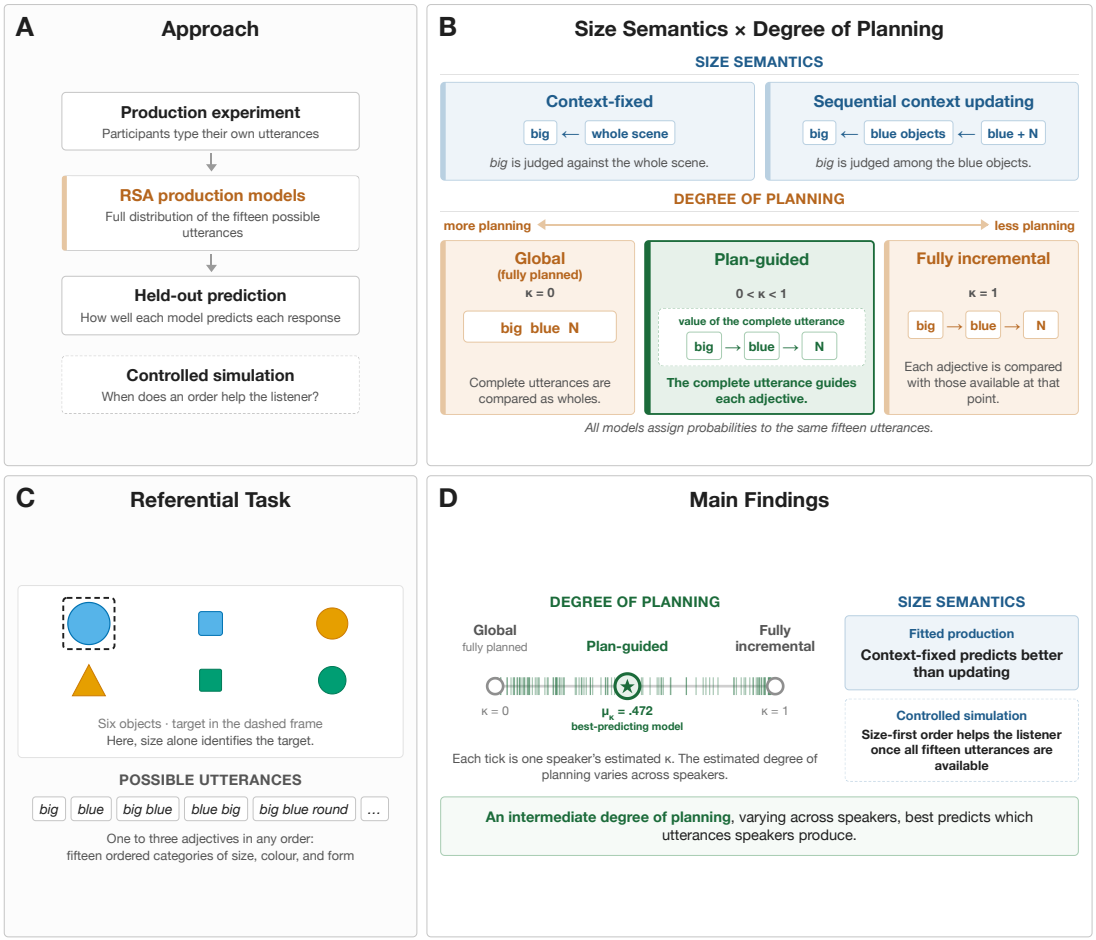


Figure 1: Overview of the approach, models, and findings. (A) RSA production models are fitted to the production experiment and compared by held-out prediction. Controlled simulations examine under which conditions hierarchical composition can favour size-first order for the listener. (B) The production models cross two size semantics with three degrees of planning. With context-fixed semantics, *big* is interpreted relative to the whole display. With sequential context updating, as proposed by hierarchical composition, the noun-adjacent adjective first narrows the comparison class for *big*. Production ranges from fully planned (global, $\kappa = 0$) through plan-guided ($0 < \kappa < 1$) to fully incremental ($\kappa = 1$). (C) An example display in which size alone identifies the target, with examples of possible utterances. (D) Plan-guided production predicts the production data best (population mean $\mu_\kappa = .472$); ticks show each participant's estimated κ. Context-fixed semantics predicts better than updating. In the simulations, updating shifted the size-first preference only slightly, and size-first order helped the listener with both size semantics once all fifteen utterances were available (Section 5).

2 Production Experiment

Whether speakers resolve the competition between size-first and colour-first order by evaluating complete utterances or single adjectives becomes more directly visible when they produce the adjectives themselves. The experiment builds on the slider-rating study of Schlotterbeck and Wang (2023), in which participants compared two phrases that differed only in the order of the size and colour adjectives, such as *den großen blauen Aufkleber* and *den blauen großen Aufkleber*. A follow-up slider study with newly generated items and a new sample replicated the relation between referential context and adjective-order preference (Appendix A.1). Size-first ratings were strongest when size alone identified the target and weakest when colour alone did. A small size-first preference remained even in colour-sufficient displays, consistent with a conventional size-first order that the production models include as a stable order term (Section 3.2). The present experiment used similar displays and manipulations, but replaced the comparison of the two phrases with direct production. Direct production allows adjective selection and utterance length to vary along with order. As in the slider study, the experiment manipulated referential context within subjects and size discriminability between subjects. Referential context varies whether size, colour, or both are needed to identify the target, manipulating the relative usefulness of the two properties (Fukumura, 2018; Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). Size discriminability separately varies how clearly size differences can be perceived. Crossing these factors allows us to ask whether the influence of size discriminability depends on which properties support reference, and how these pressures affect adjective selection, length, and order.

2.1 Method

2.1.1 Participants

The experiment was conducted online via Prolific, with compensation set slightly above the platform's median level. Participants reported German as their native language. The retained sample comprised 113 participants, 60 in the low- and 53 in the high-discriminability group. Exclusion criteria and the trials retained for each analysis appear in Table 2.

2.1.2 Materials

The task was presented as a reference game about collectible stickers, in which participants described the framed sticker so that a partner could identify it. The partner saw the same stickers in a different order and without the target frame, so spatial descriptions were uninformative. Successful descriptions therefore had to distinguish the target from the other stickers by its size, colour, or form. The original German instructions and an English translation appear in Appendix A.4. On each critical trial, participants saw six objects with one framed target (Figure 2, Panel A). The noun was always *Aufkleber* ('sticker'), and the materials varied the objects' size, colour, and form across counterbalanced lists. The target was always the largest object of its colour and form, so it could always be described as *big*, although it was not necessarily the only big object. Stimulus construction, list counterbalancing, and trial structure appear in Appendix A.

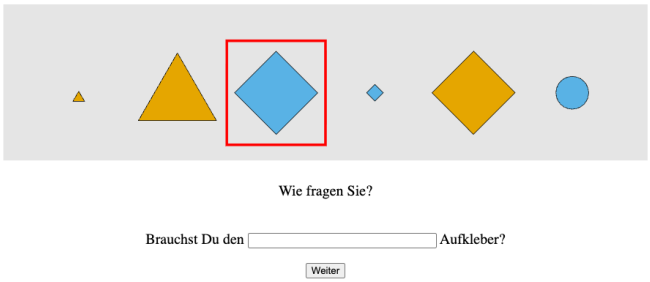
2.1.3 Design

The experiment used a 3 (referential context: first property sufficient, second property sufficient, both necessary; within-subjects) $\times$ 2 (size discriminability: high, low; between-subjects) mixed design. Referential context changed which properties were needed to identify the target, for example size alone (size sufficient), colour alone (colour sufficient), or both properties (both necessary). Target–competitor size differences were more pronounced in the high-discriminability group than in the low-discriminability group (Figure 2). The two experimental hypotheses concern the interaction between the two factors. The first predicts that lower size discriminability increases redundant adjective use more strongly when size alone is sufficient, because a redundant colour adjective can help the listener when size differences are hard to perceive (Rubio-Fernández, 2016; Degen et al., 2020). The second concerns size-initial responses when both properties are necessary, where the two efficiency accounts predict opposite effects of size discriminability. Under hierarchical composition, lower discriminability makes it less certain whether the target counts as big, which should favour placing size first, further from the noun (Scontras, Degen, and Goodman, 2017; Franke, Scontras, and Simonic, 2019). Under incremental pragmatics, higher discriminability makes size the more discriminating property, which should favour placing it first so that it narrows down the referent early (Fukumura, 2018; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). The first hypothesis thus concerns adjective selection, and the second adjective ordering. Both aspects determine whether size-first

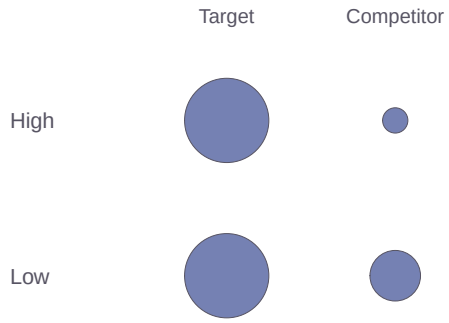
and colour-first order compete in production. The production models in Section 3 aim to reproduce these patterns along with the full distribution of utterances.

Each context was realised with three adjective pairs (size–colour, size–form, colour–form), which determined the two properties that could distinguish the target. Size and colour are the classic pair in the ordering literature and our motivating example, and the hypothesis-driven behavioural analyses focus on the size–colour trials. Size–form and colour–form trials varied which two properties were relevant from trial to trial, so that size and colour were not always the properties that distinguished the target. The inclusion of form also extends the comparison to utterances in which several properties can compete for mention and initial position. Responses from all three adjective pairs were coded with the same scheme, described below.

A



B



C

Design dimension	Levels
Referential context	First property sufficient; second property sufficient; both necessary (within-subjects)
Size discriminability	High; low (between-subjects)
Adjective pair	Size–colour; size–form; colour–form (stimulus sets, crossed with referential context)

Figure 2: Experimental design and materials. Panel A shows a trial screenshot from the production experiment: the framed object is the target, and participants typed the adjective material into the sentence frame. Panel B illustrates the size-discriminability manipulation schematically using recorded stimulus levels: the target–competitor size gap is larger under high discriminability and smaller under low discriminability. Panel C summarises the 3×2 design and the three adjective pairs.

2.1.4 Procedure

Participants saw the framed target together with the fixed sentence frame *den ___ Aufkleber* and typed the adjective material before the noun. This response format allowed us to observe which properties they mentioned, how many adjectives they used, and how they ordered them. Practice trials introduced the relevant colour and form vocabulary, and main-trial order was randomised within blocks.

The typed responses were annotated following guidelines that preserve adjective content and order. A production response could contain a single adjective, such as *blue*; two adjectives, such as *big blue*; or three, such as *big blue round*. Coding each adjective as size (S), colour (C), or form (F) gives fifteen ordered categories: three single adjectives, six two-adjective sequences, and six three-adjective sequences. All production models assign probabilities to these fifteen categories. The complete annotation scheme, including the treatment of responses outside these categories, appears in Appendix A.

For the two experimental hypotheses, the annotated responses were further classified into two measures. *Redundant adjective use* indicates a response containing more properties than needed for unique reference: multiple adjectives in a one-property-sufficient context, or three adjectives when both properties are needed. *Size-initiality* indicates that a response with more than one adjective begins with size.

Both hypotheses were tested with Bayesian hierarchical logistic models of these measures, with referential context, size discriminability, and their interaction as predictors. The models included participant intercepts, participant slopes for referential context, and item intercepts. Full specifications and the planned contrasts appear in Appendix A.

2.2 Results

Panel A of Figure 3 shows the full distribution of size–colour responses. The observed response distribution indicated that referential context shaped both adjective selection and adjective ordering. Both-necessary contexts generally elicited longer utterances, whereas one-property-sufficient contexts more often elicited a single adjective. Production also revealed substantial use of three-adjective utterances, which mention all three properties and are therefore maximally redundant. Any utterance with more than one adjective involves an ordering, and such utterances occurred mainly in size-sufficient and both-necessary contexts. When speakers mentioned both size and colour, they mostly placed size before colour, following the robust size-first convention. Utterances with colour before size, such as *colour–size–form*, were rare. The two accounts favour competing orders in colour-sufficient contexts, where hierarchical composition can favour size-first and incremental pragmatics colour-first order. Speakers, however, commonly resolved these contexts with colour alone, so the size–colour orderings that could distinguish the two accounts were rare. In this task, adjective order therefore depends on adjective selection. The two hypothesis measures below take up both aspects, with redundant adjective use indicating which adjectives speakers selected and size-initiality indicating which adjective they placed first.

Redundant adjective use. When size alone identified the target, most responses still contained more than one adjective (Figure 3, Panel B). Such redundant utterances were more frequent under low size discriminability, that is, when the target stood out less clearly as big relative to its comparison class (23.4 percentage points, 95% CrI $[15.1, 31.8]$; posterior probability $> .999$). This increase was 15.3 percentage points larger than the average effect of low discriminability in the both-necessary and colour-sufficient contexts (95% CrI $[7.2, 23.6]$; posterior probability $> .999$). This pattern supports the first hypothesis, which predicted that lower size discriminability would increase redundant adjective use particularly when size alone was sufficient.

Initial adjective. Among responses with more than one adjective, speakers began with size most often when size alone identified the target, less often when both properties were needed, and least often when colour alone did (Figure 3, Panel C). Both differences were credible (18.0 percentage points, 95% CrI $[12.6, 23.4]$; 47.9 percentage points, 95% CrI $[39.6, 55.6]$). Size discriminability mattered mainly when both properties were needed, where speakers began with size more often under high size discriminability (14.4 percentage points, 95% CrI $[4.3, 24.4]$; posterior probability .997). This increase was 15.9 percentage points larger than the average effect of high discriminability in the two one-property-sufficient contexts (95% CrI $[6.7, 25.1]$; posterior probability $> .999$). Hierarchical composition predicted more size-initial responses under low discriminability, when it is less certain whether the target counts as big, whereas incremental pragmatics predicted more under high discriminability, when size better narrows down the referent. The observed increase under high discriminability is therefore consistent with incremental pragmatics and opposite to the prediction of hierarchical composition.

2.3 Discussion

The production data suggest that referential context and size discriminability affected both which adjectives speakers selected and which adjective they placed first, with the discriminability effect on order in the direction predicted by incremental pragmatics. Beyond these two measures, three-adjective responses were common whenever size was relevant, making up about a third of responses when size alone was sufficient or both properties were needed. These responses mention every property of the target and therefore also fix the order of three adjectives at once. Such redundancy need not reflect a failure to be informative, as Rubio-Fernandez (2019, p. 7) proposed that speakers who share a display with the listener can rely on visual heuristics at a low processing cost, such as the expectation that a denser display makes the listener's visual search harder. Describing the target completely could serve as such a heuristic in the present six-object displays, helping the listener find the referent without the speaker determining which properties are strictly needed. Speakers also differed considerably in how much they said, from speakers who rarely produced three adjectives to speakers who did so in most trials.

A production model therefore needs to explain redundant responses alongside minimal ones, including descriptions that mention all three properties. The model also needs to let adjective order depend on adjective selection, so that discriminability can change which property speakers produce first. The variation across speakers further suggests that speakers may differ in how far they evaluate the complete description before producing it. The following modelling framework treats adjective selection and ordering within a unified account, allowing us to compare how hierarchical composition and the degree of planning contribute to speakers' descriptions.

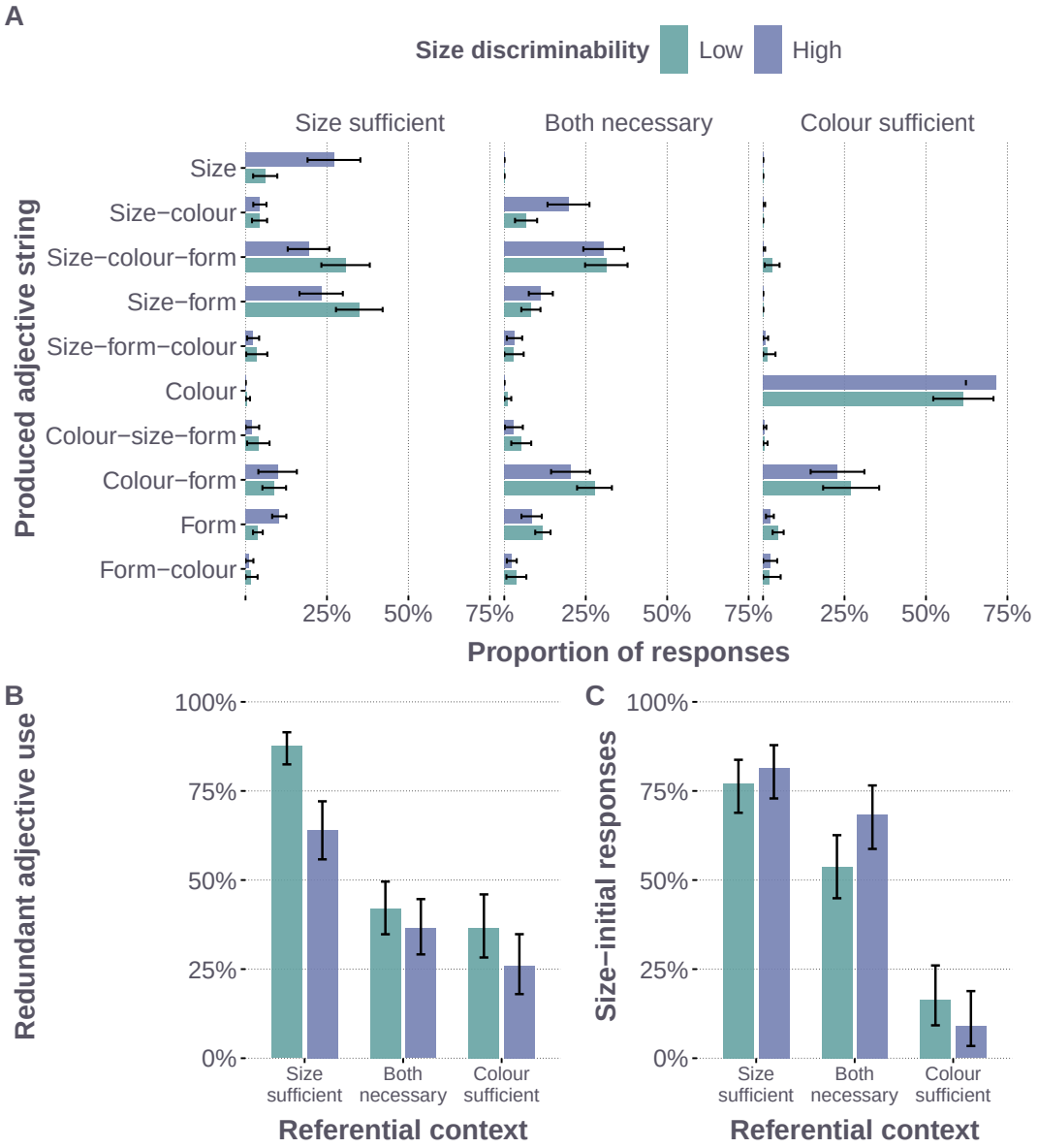


Figure 3: Size–colour production responses by referential context and size discriminability. Panel A shows the ordered response distribution, with bars for mean participant-level proportions and 95% confidence intervals. Adjective labels preserve surface order, and the panel shows responses reaching at least 2% in one condition. Panel B shows redundant adjective use, defined as a multi-adjective response in a one-property-sufficient context or a three-adjective response when both properties are necessary. Panel C shows the probability that a response with more than one adjective begins with size. In Panels B and C, bars show Bayesian posterior medians and error bars show 95% credible intervals.

3 Production Models

In the production data, the context affected which adjectives speakers mentioned, how many they used, and which adjective they placed first. These choices depend on how much of the utterance a speaker evaluates before producing it. In a colour-sufficient context, where *blue* alone identifies the target, a speaker who evaluates complete utterances compares alternatives such as *blue*, *big blue*, and *blue big*, whereas a speaker who evaluates one adjective at a time considers the first adjective on its own. Both speakers pursue the same goal of identifying the target; they differ only in how far ahead they look. Speakers who adapted both the content and the first adjective to the context while keeping the conventional order may lie between these two ends, evaluating each adjective while looking ahead to the complete utterance. We therefore model a global speaker, a fully incremental speaker, and a plan-guided speaker between them.

The Rational Speech Act (RSA) framework models a speaker who selects utterances according to their informativeness about the intended referent and their production cost (Frank and Goodman, 2012; Goodman and Frank, 2016). Standard RSA speakers evaluate complete utterances, whereas incremental variants evaluate one word at a time (Cohn-Gordon, Goodman, and Potts, 2019; Schlotterbeck and Wang, 2023). Our model keeps this measure of informativeness but treats the choice between these two forms of evaluation as a matter of degree, estimated from the data. Planning cannot be observed directly in typed utterances, so the model infers it from the distribution of utterances that speakers produce. The models also differ in whether the comparison class for size is updated as adjective meanings compose from the noun outwards (Figure 1, Panel B).

3.1 State Space, Utterance Space, and Semantic Regimes

The models are defined over the displays and descriptions of the production experiment. The states are the six objects of a display, represented symbolically by the recorded size value, the colour category, and the form category of each object, with the target as the intended referent. The utterances are the fifteen adjective strings coded in the experiment, which differ in which of the adjectives for size, colour, and form they contain and in which order (Section 2).

Size has a threshold-based meaning, under which an object counts as *big* to the degree that its size exceeds a threshold set by its comparison class (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019). Colour has a relaxed meaning, under which objects of the named colour receive most, but not all, of the weight (Degen et al., 2020). Form is treated like colour, as both properties are less subjective than size (Scontras, Degen, and Goodman, 2017, p. 56).

For every complete or partial adjective string, the literal interpretation L composes the adjectives hierarchically, applying modification from the noun outwards. L corresponds to the literal listener of RSA models and serves here as a semantic layer and not as a separate agent. For example, in *big blue sticker*, *blue* modifies *sticker* before *big* is interpreted. Let $u = w_1 \dots w_T$ be an utterance, so that w_T is the adjective next to the noun. The context C comprises the six objects o of the display, among which o^* is the target. Starting from a uniform prior over the objects in C, $L_{T+1} = P_0$, each adjective updates the interpretation from the noun outwards:

$$L_t(o \mid u, C) \propto L_{t+1}(o \mid u, C)\, [\![w_t]\!]_{C_t}(o), \qquad t = T, \dots, 1, \tag{1}$$

where $[\![w_t]\!]_{C_t}$ is the meaning of w_t given its comparison class C_t, the context against which it is interpreted. Each step thus takes the distribution produced by the inner modifiers as the prior for the next, outer modifier. The final state L_1 gives the literal interpretation of the utterance as a distribution over objects.

Hierarchical composition affects order only if an inner adjective can change how an outer adjective is interpreted. The two semantic regimes differ in exactly this respect. Under *context-fixed* semantics, the comparison class for size remains the whole display, $C_t = C$. Under *sequential context updating*, the distribution produced by the inner adjectives becomes the prior of the comparison class for size, $C_t = L_{t+1}$. Consider a display in which the target is a medium-sized blue object, the other blue objects are small, and the red objects are larger than the target. Against the whole display, the target is not particularly big, so *big* applies more strongly to the red objects. In *big blue*, however, *blue* is interpreted first, and *big* is then judged among objects weighted by how well they match *blue*. Among the blue objects the target is the largest, so that *big* now singles it out. Colour meanings are relaxed, so the red objects still count as blue to a small degree and this restriction remains partial. In *blue big*, by contrast, *big* is next to the noun and is interpreted against the whole display. Updating can therefore make the size-first order *big blue* more informative than *blue big*, whereas under context-fixed semantics both orders receive the same interpretation. The details of threshold construction and semantic parameters appear in Appendices B and F.

3.2 Production Architectures

The literal interpretation implements hierarchical composition, applying the adjectives from the noun outwards. Production, by contrast, proceeds from left to right, and incremental pragmatics values each adjective by how much it narrows down the referent at its position in this sequence. The speaker therefore evaluates the information available after each adjective in the surface sequence.

We compare three production architectures, a global, a fully incremental and a plan-guided speaker. All three share a *prefix utility*, which accumulates how well the adjectives in an utterance identify the target at each adjective position. This accumulation means that an utterance gains value when it identifies the target early, not only when it is complete. The production architectures differ in how this utility is compared with alternatives, across complete utterances or among the adjectives available at each point. Let $w_{1:t} = w_1 \dots w_t$ be the prefix comprising the first t adjectives. At position $t+1$, the utility of producing adjective w_{t+1} after the prefix $w_{1:t}$ is

$$U(w_{t+1} \mid w_{1:t}, o^*, C) = \alpha \log L(o^* \mid w_{1:t+1}, C) + \lambda_{\text{salience}} z_{w_{t+1}}, \tag{2}$$

where the pragmatic-optimality parameter α controls sensitivity to informativeness and $z_{w_{t+1}}$ represents the visual salience of the candidate property. Visually salient properties, colour in particular, are often mentioned even when they are redundant (Rubio-Fernández, 2016, p. 2), and more salient properties tend to be produced earlier (Fukumura, 2018, p. 47). The salience score is higher for colour than for size and increases when a property contrasts with the other objects in the display (Appendix B).

Like the standard RSA speaker (Frank and Goodman, 2012; Goodman and Frank, 2016), the *global* speaker compares complete utterances,

$$S_{\text{G}}(u \mid o^*, C) \propto \exp\Big\{ \sum_{t=0}^{T-1} U(w_{t+1} \mid w_{1:t}) + \beta\Omega(u) + \Lambda(u, C) \Big\}, \tag{3}$$

where $U(w_{t+1} \mid w_{1:t})$ abbreviates $U(w_{t+1} \mid w_{1:t}, o^*, C)$.[2]

Following common practice in Bayesian cognitive modelling, we include auxiliary components that link the model to the task (Lee and Wagenmakers, 2013; Franke, 2016). Our account does not derive these components, yet a model that ignored them could misattribute their effects to the degree of planning. The order term $\beta\Omega(u)$ weights a fixed order score from a German language model. Schlotterbeck and Wang (2023, p. 124) found a strong preference for size-first order, and our replication found this preference even when colour alone identified the target (Appendix A.1). This stable order term reflects conventions learned from language use, which may themselves have been shaped by efficiency pressures over time (Franke, Scontras, and Simonic, 2019). The length and selection terms $\Lambda(u, C)$ take the place of the utterance cost in standard RSA. The parameters of these components are estimated jointly with those of each architecture, so uncertainty about them is carried into the inference about planning. These components are identical across architectures, so that the architectures differ only in how utterances are evaluated. Their specification appears in Appendix B.

The *fully incremental* speaker uses the same utilities, order term, and length and selection terms as the global speaker, but compares each adjective only with the other adjectives still available after the prefix,

$$S_{\text{I}}(u \mid o^*, C) \propto \exp\big\{\beta\Omega(u) + \Lambda(u, C)\big\} \prod_{t=0}^{T-1} \frac{\exp U(w_{t+1} \mid w_{1:t})}{\sum_{w' \in \mathcal{A}(w_{1:t})} \exp U(w' \mid w_{1:t})}, \tag{4}$$

where $\mathcal{A}(w_{1:t})$ contains the adjective types still available after the prefix. Each fraction gives the probability of producing w_{t+1} after the prefix $w_{1:t}$, relative to the other adjectives available at that point. The two speakers differ in whether the adjectives that follow can compensate for a first adjective whose utility is low relative to its competitors. Under the global speaker, only the summed utility of the complete utterance counts, so high utilities later in the utterance can offset a low utility at the first position. Under the fully incremental speaker, the low utility lowers the probability of the first fraction, and each later fraction is at most one, so later adjectives cannot offset it. The same holds at each later position, and the fully incremental speaker therefore does not look ahead. Two utterances with the same summed utility can thus receive different probabilities, depending on how strongly each of their adjectives competes with the alternatives at its position.

The *plan-guided* speaker combines the global and the fully incremental speaker as a geometric interpolation,

$$S_\kappa(u \mid o^*, C) \propto S_{\text{G}}(u \mid o^*, C)^{1-\kappa}\, S_{\text{I}}(u \mid o^*, C)^{\kappa}, \tag{5}$$

[2] Unlike in standard RSA, the utility of the global speaker here is summed over the adjectives of an utterance, which predicted the production data better (Appendix C.5).

which is renormalised over the fifteen completed utterances. With $\kappa = 0$, it reduces to the global speaker, and with $\kappa = 1$ to the fully incremental speaker.[3] Consider a context in which colour alone identifies the target. Under the global speaker, the prefix *big*, which does not identify the target, is one term in the summed utility of *big blue*, which is compared with the other complete utterances. Under the fully incremental speaker, *big* must first win a separate competition at the first position against *blue*, which identifies the target on its own. At intermediate values of κ, the utility of *blue* after *big* can offset the low utility of *big* at the first position only in part, and less so the higher κ. The log-odds of *big blue* over *blue big* move linearly from their global value at $\kappa = 0$ to their fully incremental value at $\kappa = 1$. An intermediate κ thus allows the speaker to keep the conventional order *big blue*, while the competition from *blue* at the first position makes *blue big* more likely than under global production. Lower values of κ therefore correspond to a greater degree of planning.[4]

4 Model Comparison

The production models differ in two assumptions, each derived from one of the two efficiency accounts. The degree of planning κ determines whether the speaker evaluates the complete utterance or each adjective as it is produced, the scope of evaluation that separates the two accounts. The semantic regime determines whether the comparison class for size narrows as adjective meanings compose from the noun outwards, as proposed by the hierarchical compositional account. The model comparison tests how each assumption contributes to the observed distribution of utterances, while holding all other components constant. If speakers reconcile the two pressures by evaluating each adjective while looking ahead to the complete utterance, an intermediate degree of planning should predict the utterances better than both endpoints. We first compare the production architectures and the semantic regimes, then ask whether the degree of planning varies across speakers, and finally examine which production patterns the best-predicting model captures.

4.1 Method

To isolate the two assumptions, we fitted models that differ only in production architecture and semantic regime. Crossing the three production architectures with the two semantic regimes gives a 3×2 factorial design of six models (Figure 1, Panel B), which share all other components. The design estimates the predictive contribution of each assumption and their interaction. The global and fully incremental models fix κ at zero and one, and the plan-guided model estimates one κ shared across participants. Two further models refine the plan-guided architecture with participant-specific κ_i under each semantic regime, so the main comparisons rest on eight fitted models. All models also allow participant variation in pragmatic optimality α_i and stable-order weight β_i.

We fitted the eight models as hierarchical Bayesian models with the structure shown in Figure 4. For participant i on trial n, the model maps the display C_{in} and the stable-order score $\Omega(u)$ to a probability $\pi_{in}(u)$ over the fifteen utterances, given by the speaker of the respective architecture (Equation (5)) with Laplace smoothing (Appendix B). The likelihood of an observed response is this probability, $p(y_{in} = u \mid C_{in}) = \pi_{in}(u)$, so that each response is treated as a draw from a categorical distribution over the fifteen utterances. Participant-specific parameters are drawn from population distributions, so the estimate for each participant is informed by the responses of all participants. Bayesian inference was performed in NumPyro (Phan, Pradhan, and Jankowiak, 2019), and posterior distributions were sampled with the No-U-Turn Sampler (Hoffman and Gelman, 2014).

Bayesian models are commonly compared either by Bayes factors, which evaluate how well each model predicts the observed data under its prior, or by how well the fitted models predict held-out data. We use the second approach and compare models by their approximate leave-one-out expected log predictive density (ELPD), which measures how much probability a model assigns to responses held out from fitting (Vehtari, Gelman, and Gabry, 2017). Higher ELPD indicates better prediction of the complete utterance distribution. Additional parameters raise ELPD only if they improve the prediction of held-out responses, so model complexity is taken into account. For each pair of models, we report the difference in ELPD (ΔELPD) with its 95% credible interval, and an interval that excludes zero indicates a credible difference in predictive accuracy. Robustness checks refitted the models with alternative auxiliary components or evaluated them on new participants, to test whether the conclusions depend on these choices (Appendix C.5). The model specification, inference procedure, numerical diagnostics, and posterior parameter estimates appear in Appendices B, C and C.2.

[3] This contrast corresponds to that between locally and globally normalised sequence models in natural language processing (Lafferty, McCallum, and Pereira, 2001; Andor et al., 2016).

[4] The parameter κ plays a role similar to the future discount in the incremental production model of Futrell (2023), which determines how much the value of later words contributes to the production of the current word.

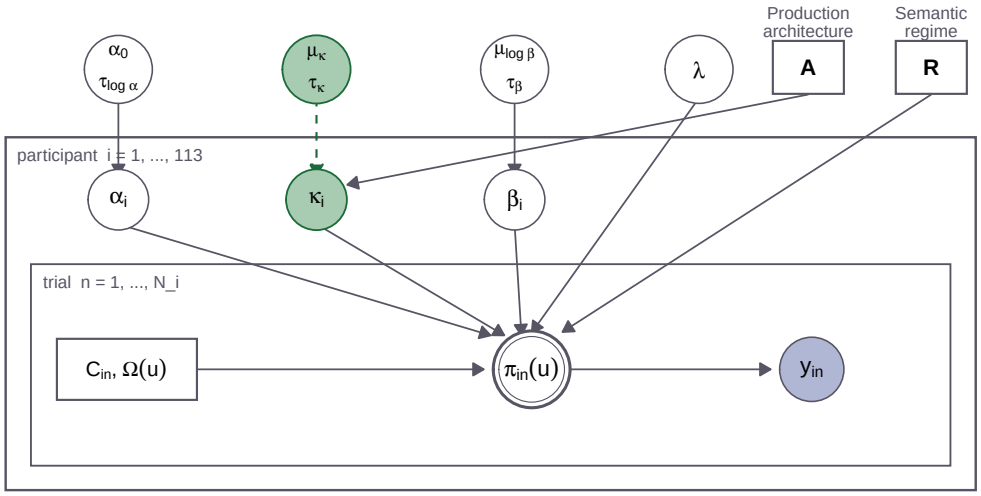


Figure 4: Hierarchical structure of the production models. Solid circles represent estimated quantities, squares represent fixed model or trial inputs, the double circle represents the deterministic utterance distribution, and the shaded circle represents the observed response. The fixed variables A and R index the production architecture and the semantic regime. For participant i on trial $n = 1, \ldots, N_i$, the display C_{in} and the stable-order score $\Omega(u)$ determine the probability $\pi_{in}(u)$ of each utterance u, from which the response y_{in} is drawn. The participant parameters are pragmatic optimality α_i, which scales the weight of informativeness, the stable-order weight β_i, which scales the stable-order score, and the degree of planning κ_i. They are drawn from population distributions with locations α_0, $\mu_{\log \beta}$, and μ_κ and scales $\tau_{\log \alpha}$, τ_β, and τ_κ. The vector $\boldsymbol{\lambda}$ collects the coefficients shared across participants, which weight salience and the length and selection terms. The dashed κ hierarchy applies only to the plan-guided models with participant-specific κ_i; the other plan-guided models estimate one shared κ, and the global and fully incremental models fix κ at zero and one.

4.2 Results

We begin with the interaction between production architecture and semantic regime. The predictive cost of sequential context updating differed across production architectures, and it was largest for plan-guided and smallest for fully incremental production (Figure 5, Panel C; plan-guided minus fully incremental production in the updating-minus-fixed contrast: $\Delta\text{ELPD} = -37.9$, 95% CrI $[-55.0, -21.6]$). In the model, context updating changes only the interpretation of prefixes with more than one adjective, as the innermost adjective is always interpreted against the whole display. The fully incremental speaker selects the first adjective among single-adjective prefixes, whose interpretation is the same under both semantic regimes, and is therefore less affected by the semantic regime than speakers who also evaluate complete utterances. The interaction did not reverse either factor, as plan-guided production predicted best under both semantic regimes and context-fixed semantics predicted better under all three architectures. We therefore report the main effect of each factor averaged across the levels of the other, and the contrasts within each level appear in Appendix C.

Production architecture. We then asked which degree of planning best predicts the utterances speakers produced. The Bayesian model comparison favoured plan-guided production over both global and fully incremental production (Figure 5, Panel A). Averaged across semantic regimes, the plan-guided model had higher predictive accuracy than the global model ($\Delta\text{ELPD} = 256.9$, 95% CrI $[73.8, 405.4]$) and the fully incremental model ($\Delta\text{ELPD} = 548.5$, 95% CrI $[348.5, 793.5]$). A robustness check without the condition-specific length and selection terms led to the same result (Appendix C.5). A robustness check that valued utterances only by the target probability after the completed utterance led to the same result (Appendix C.5). These results supported the plan-guided production hypothesis that the summed utility of the complete utterance and the competition among the adjectives available at each position jointly shape the distribution of utterances.

Hierarchical semantic composition. We next asked whether context updating, as proposed by the hierarchical composition account, improves prediction. Averaged across production architectures, context-fixed semantics pre-

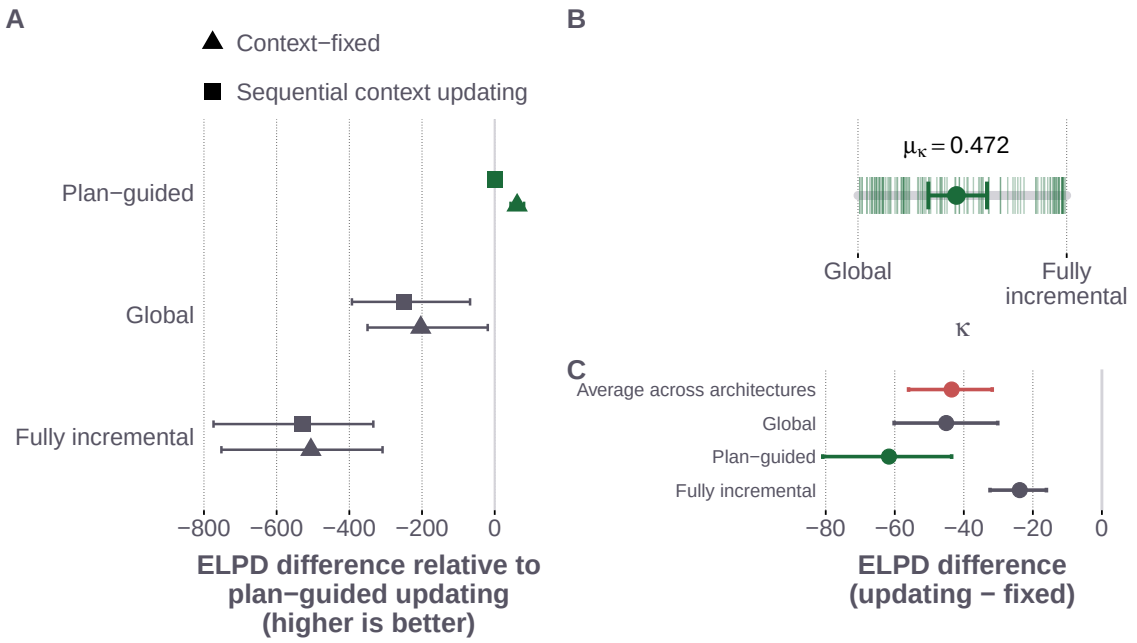


Figure 5: Matched comparisons of production architecture and semantic regime. Panel A shows the ELPD difference of each of the six matched models from the plan-guided model with sequential context updating; higher values indicate better leave-one-out prediction. Panel B shows the population mean μ_κ of the degree of planning in the best-predicting model, between the global and fully incremental endpoints, with ticks for the posterior means of the participant-specific κ_i. Panel C shows the updating-minus-fixed ELPD difference for each production architecture and their average. Intervals in Panels A and C are 95% credible intervals from the participant-level Bayesian bootstrap; Panel B shows the 95% posterior credible interval for μ_κ. The vertical line in Panel C marks zero.

dicted better than sequential context updating (updating minus fixed $\Delta\text{ELPD} = -43.5$, 95% CrI $[-56.0, -31.8]$; Panel C). A robustness check in which colour adjectives distinguished more sharply between matching and non-matching objects led to the same result (Appendix C.5). Context updating thus did not improve the prediction of the utterances.

Best-predicting model. Beyond these comparisons, one may ask whether the model also describes the data well. To develop such a model, we further took heterogeneity across participants into account by allowing the degree of planning to vary across them. Participant-specific κ_i improved prediction further ($\Delta\text{ELPD} = 364.7$, 95% CrI $[248.2, 521.0]$), and context-fixed semantics again predicted better than sequential context updating ($\Delta\text{ELPD} = -54.1$, 95% CrI $[-69.0, -39.7]$). The best-predicting model was therefore the plan-guided model with context-fixed semantics and participant-specific κ_i. In this model, the population mean of κ_i lay between the global and fully incremental endpoints ($\mu_\kappa = .472$, 95% CrI $[.337, .618]$), and the participants varied widely around it (Figure 5, Panel B). Two robustness checks showed that this variation in κ_i does not reduce to a mixture of global and fully incremental speakers, and that all comparisons led to the same conclusions when held-out participants were predicted as new speakers (Appendix C.5).

Posterior predictive checks compared this model with the observed responses (Figure 6). The model described the main patterns of the production data well and reproduced the effects tested by both experimental hypotheses in the observed direction. Panel A shows the two hypothesis measures from the production experiment, redundant adjective use and size-initial responses, for size–colour trials in each referential context, and Panel B shows the proportion of each utterance in each experimental condition. The model captured how referential context shaped redundant adjective use and size-initial responses, as well as the larger increase in redundant adjective use under low size discriminability when size alone was sufficient (Panel A). In both-necessary displays, the model also predicted more size-initial responses under high size discriminability, but the predicted increase was smaller than the observed one (6 versus 14 percentage points). The model represents the sizes of the objects but not how clearly their difference can be perceived, which may explain why it underestimated this effect of size discriminability. When colour alone was sufficient, the model also placed size first in multi-adjective responses more often than

speakers did, as the same size semantics leave *big* weakly informative even for a target that is not the largest object. Across all 270 response cells, the predicted proportions correlated with the observed proportions at $r = .909$, with a mean absolute error of 3.0 percentage points (Panel B).

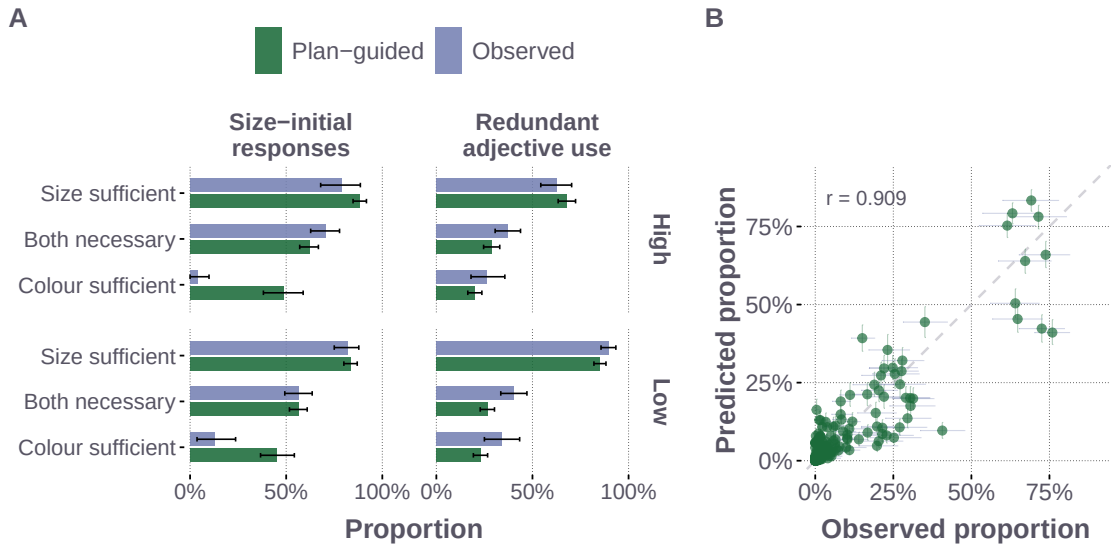


Figure 6: Posterior-predictive adequacy of the best-predicting plan-guided model. Panel A compares observed and posterior-predicted size-initial responses (among responses with more than one adjective) and redundant adjective use in size–colour trials, with referential context on the vertical axis and size discriminability in facet rows. Intervals show participant-bootstrap 95% intervals for the observed proportions and 95% posterior-predictive intervals for the model predictions. Panel B compares all 270 response cells; horizontal intervals show participant-bootstrap uncertainty in the observations, vertical intervals show 95% posterior-predictive uncertainty, and the dashed line marks equality.

4.3 Discussion

The predictive evidence thus locates the resolution of the competition in how far ahead utterances are evaluated. An intermediate degree of planning, which varied across speakers, predicted the utterances best, whereas context updating did not improve prediction.

We further analysed the model predictions to locate where these predictive differences arose. Most of the plan-guided advantage came from responses that began with colour (81.7% of the advantage over the average of the global and fully incremental models). Colour is more salient than size and often identifies the target on its own, so it competes strongly for the first position. The global speaker values a colour-initial utterance only through the summed utility of the complete utterance, whereas the plan-guided speaker also lets colour compete with the other adjectives at the first position, as in the example of Section 3.2. In most conditions, this competition brought the predicted proportion of colour-initial responses closer to how often speakers began with colour.

A closer look at the semantic contrast, on the other hand, showed that context-fixed semantics predicted better how many adjectives speakers produced ($\Delta\text{ELPD} = 21.5$, 95% CrI $[12.4, 30.9]$) and which properties they selected ($\Delta\text{ELPD} = 31.1$, 95% CrI $[23.5, 39.2]$), but not the order of the selected adjectives ($\Delta\text{ELPD} = 1.5$, 95% CrI $[-1.8, 4.9]$). Hierarchical composition has been proposed mainly as an explanation of adjective order, yet it did not improve the prediction of order in these data. One possibility is that the compositional pressure has shaped the conventional order over repeated language use (Scontras, Degen, and Goodman, 2019), so that its effect is carried by the stable order term instead of being recomputed for each utterance. Consistent with this possibility, context updating improved prediction when the stable order term was removed ($\Delta\text{ELPD} = 23.5$, 95% CrI $[11.7, 37.5]$). Another possibility is that the communicative advantage of size-first order is small and depends on the context, so that it need not appear in the limited set of displays used in the experiment. We therefore turn to a simulation to investigate each of these two possibilities further.

5 Simulation

The predictive model comparison presents a complex picture for the hierarchical composition account. Contrary to our prediction based on this account, context updating predicted the utterances worse than context-fixed semantics, but this disadvantage concerned which and how many adjectives speakers produced, and not their order. The proposed benefit of hierarchical composition, by contrast, concerns order, as even a small advantage for the listener across the many contexts of language use could make size-first order conventional over time (Scontras, Degen, and Goodman, 2019, p. 7:19). Such an advantage need not appear in the displays of a single experiment, and the production data do not measure it, as it concerns how well a listener identifies the referent. Earlier simulations found such an advantage when two complete orders were compared with respect to communicative success, which they defined as the probability that a listener identifies the intended referent from the meaning of the order (Franke, Scontras, and Simonic, 2019). However, these simulations considered no alternatives other than the two orders, and they approximated communication by presupposing the speaker and the listener that this definition requires. Our production models developed here can fill this gap by providing such a speaker, who chooses among utterances of different lengths, and they were evaluated against the utterances that speakers produced. Furthermore, we add a pragmatic listener who reasons about the speaker's choices, which yields the model of communication assumed in this definition. We use this simulation setting to identify the conditions under which context updating gives size-first order a communicative advantage across many randomly generated contexts, when both hierarchical composition and incremental pragmatics shape the speaker's choices and the speaker can choose among more alternatives than the two orders. This allows us to ask what each mechanism can do in principle, independently of the fit to particular data, as the simulation retains the assumptions of the two accounts and omits the auxiliary components that the production models include to link them to the task.

5.1 Method

Design. The contexts followed the design of the production experiment, with changes that extend it beyond the experimental items. As in the experiment, size discriminability varies how clearly the target stands out as big relative to its comparison class. Size discriminability has three levels here, low, standard, and high, which correspond to a narrow, an intermediate, and a wide spread of object sizes. Unlike in the experiment, a context contains 2 to 30 objects instead of six, the target is always the largest object and is blue, and the colours of the other objects were assigned at random instead of by referential context. These changes cover many contexts beyond the experimental items, in which colour restricts the comparison class for size to different degrees, as the number of blue competitors varies. The upper part of Table 1 lists the levels of the two context factors. For each of their 24 combinations, we generated $1{,}000$ contexts with random object sizes and colours, so that the results do not depend on a particular arrangement of objects.

We first tried to reproduce the earlier finding by implementing a global speaker as closely as possible to the original simulation study, which compared the two orders (Franke, Scontras, and Simonic, 2019). We then evaluated the six speaker models of the production model comparison, which cross production architecture with semantic regime (lower part of Table 1). Each model was evaluated with three sets of alternatives, which extend step by step from the two orders of the earlier simulations to the fifteen utterances of the production models. With the two orders, the global speaker is the one implemented for this reproduction. The intermediate set adds *big* alone and *blue* alone, as speakers in the production experiment often identified the target with a single adjective when one property sufficed. Whether such alternatives favour one order or the other is not clear in advance, as a single adjective that identifies the target changes what the use of two adjectives conveys to the listener. Each model was evaluated under the semantic values of the production models and, to test whether the results depend on these values, averaged over 125 settings of a semantic parameter grid (Appendix G). In total, the simulation required about 54 million model evaluations. The full specification appears in Appendices G and G.1.

Measure. The pragmatic listener L^{prag} applies Bayes' rule to the speaker distribution, using an equal prior over objects. As the dependent measure, we computed the size-first preference, the difference in the listener's target probability after the two orders,

$$\Delta_{\text{SF}} = L^{\text{prag}}(o^* \mid \text{“big blue”}) - L^{\text{prag}}(o^* \mid \text{“blue big”}). \tag{6}$$

Positive values favour size-first order. We report the mean preference across contexts with ± 1.96 Monte Carlo standard errors, which reflect the variability due to the random sampling of contexts and are computed after averaging the semantic parameter settings within each context.

Table 1: Factors of the simulation. All factors are fully crossed. With the two orders, the global speaker was implemented as in the earlier simulations. In all speaker models, pragmatic optimality is $\alpha = 1$, and the auxiliary components and the Laplace smoothing are omitted.

Factor	Levels	Number of levels
Context generation		
Number of objects	2, 6, 10, 14, 18, 22, 26, 30	8
Size discriminability	Low, standard, high (size spread 2.0, 7.75, 15.0)	3
Speaker models		
Production architecture	Global ($\kappa = 0$), plan-guided ($\kappa = .5$), fully incremental ($\kappa = 1$)	3
Semantic regime	Context-fixed, context updating	2
Alternatives	Two orders; two orders with *big* alone and *blue* alone; all fifteen utterances	3

5.2 Results

Figure 7 shows the size-first preference of all speaker models under the three sets of alternatives, averaged over the 125 settings of the semantic parameters. Overall, the preference shifted from colour-first towards size-first order as alternatives were added and was stronger the more the speaker planned, whereas the two semantic regimes differed little.

With the global speaker and the two orders, context updating yielded a size-first preference of approximately 1.1 percentage points (± 0.04), which replicates the finding of the earlier simulations (Franke, Scontras, and Simonic, 2019). Context-fixed semantics, on the other hand, yielded no difference between the orders, as the two orders then have the same completed interpretation and identify the target equally well. This symmetry also served as a sanity check of our implementation. Context updating breaks the symmetry because colour restricts the comparison class for size.

In most other speaker models, context updating likewise shifted the preference towards size-first order, by approximately 0.5 percentage points. For fully incremental production with the two orders and with the single adjectives, context updating did not change the preference. This speaker selects the first adjective among single adjectives, whose interpretation is the same under both semantic regimes (Section 4.2). Hierarchical composition thus favoured size-first order in principle, but its contribution was small.

The production architecture determined the direction of the preference when few alternatives were available. With the two orders, plan-guided and fully incremental production favoured colour-first order, by approximately 2.2 to 2.7 percentage points, and with the single adjectives as alternatives they still did so (approximately 1.2 to 2.7 percentage points), whereas global production did not favour colour-first order. The first adjectives can differ in informativeness even when the completed meanings are the same, and the competition at the first position in plan-guided and fully incremental production favoured colour, in line with the incremental preference for placing the more discriminating adjective first. The more the speaker planned, the stronger the size-first preference, and with all fifteen utterances the preference was strongest for global production (approximately 2.6 to 3.1 percentage points), followed by plan-guided (1.0 to 1.6) and fully incremental production (0.6 to 1.1).

Adding alternatives shifted the preference towards size-first order in all three architectures, and with all fifteen utterances all three favoured size-first order. The single adjectives allow the speaker to identify the target with one adjective, which changes how the listener interprets the use of two adjectives. At the semantic values of the production models, where colour is only weakly informative, plan-guided and fully incremental production favoured size-first order already with the two orders (approximately 2.1 percentage points), all three architectures favoured it with all fifteen utterances (1.0 to 1.9), and context updating left the preference unchanged.

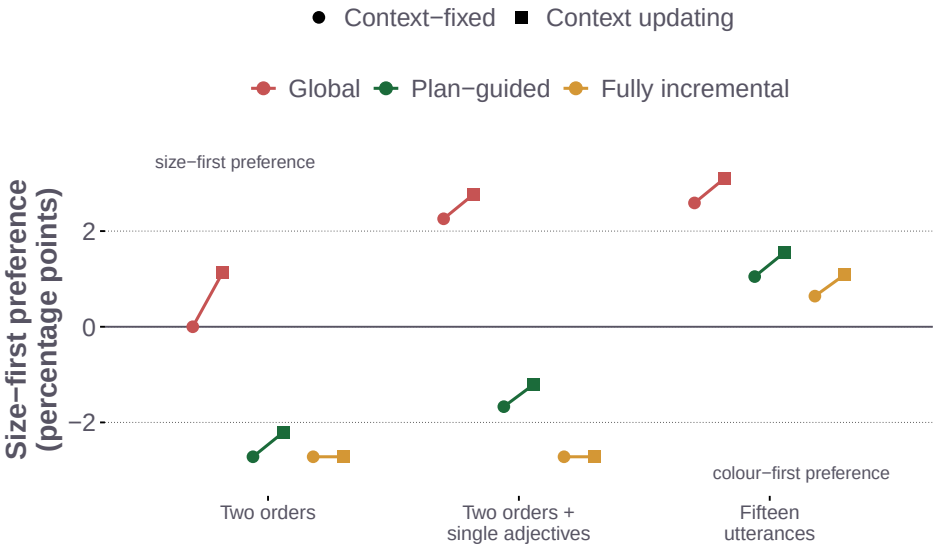


Figure 7: Size-first preference in pragmatic target recovery under the semantic parameter grid, on the same 24,000 random contexts. In the two-order condition, the global speaker reproduces the earlier simulations, and the plan-guided and fully incremental speakers are those of the production models; the other conditions show the six speaker models. Lines connect context-fixed semantics (circles) and context updating (squares) within each production architecture. Points show context-averaged target-probability differences; positive values indicate a size-first and negative values a colour-first preference. Intervals of ± 1.96 Monte Carlo standard errors are at most 0.16 percentage points and are not shown.

5.3 Discussion

In this simulation, hierarchical composition favoured size-first order in principle, but its contribution was small compared with that of the alternatives and the degree of planning, and it disappeared at the semantic values that predicted the production data. We further analysed this and found that, in the production models, the size-first preference was carried by the stable order term, consistent with the improvement of context updating when this term was removed (Section 4.3). These results suggest why context-fixed semantics predicted the production data better and how this can coexist with the theoretical result. Context updating did change which adjectives the models selected, as restricting the comparison class makes size more informative after colour or form. It thereby raised the predicted proportion of redundant three-adjective utterances, for example to 31% of size–colour–form responses in size-sufficient size–form trials with high discriminability, compared with 27% under context-fixed semantics, but speakers produced such utterances less often than either model predicted (21%).

The simulation also showed colour-first preferences, which arose for plan-guided and fully incremental production when few alternatives were available. In these speakers, colour won the competition at the first position, as incremental pragmatics predicts when colour is the more discriminating property. With all fifteen utterances, however, even these speakers favoured size-first order, under both semantic regimes and also at the semantic values of the production models. The size-first advantage for the listener thus arose from the alternatives available to the speaker, and it did not require hierarchical composition. These alternatives are the utterances that speakers produced in the experiment, and the speakers who choose among them are production models evaluated against these utterances. The simulation thus supports the proposal that communication favours size-first order across many contexts, which could make this order conventional over repeated language use (Scontras, Degen, and Goodman, 2019, p. 7:19), and it locates the source of this advantage in the alternatives that speakers produce, a source other than hierarchical composition. In the production data, speakers likewise rarely placed colour before size and often produced colour alone when colour sufficed, consistent with an incremental pressure that shows mainly in which adjectives speakers select.

6 General Discussion

The present findings link adjective ordering to how speakers plan referential utterances. We asked how speakers reconcile the pressures from hierarchical composition and incremental pragmatics when selecting and ordering ad-

jectives, and the results point to the scope of evaluation. The production data were best predicted by plan-guided production, in which each adjective is evaluated while looking ahead to the complete utterance, and the degree of planning varied across speakers. In this model, incremental pragmatics shaped which adjectives speakers mentioned and which came first. In the contexts where the two accounts favour competing orders, speakers mostly produced colour alone, so that orderings of size and colour were rare, and when both properties were needed, speakers began with size more often under high size discriminability. Hierarchical composition, by contrast, appeared to contribute through the conventional order of size and colour. When speakers mentioned both properties, they still placed size first, even when size did not help to identify the target, and context updating added no predictive benefit over this stable convention. The simulation suggests a source for such a convention. Once the speaker could choose among the utterances that speakers produced, size-first order helped the listener under every degree of planning, even without hierarchical composition. Speakers may thus reconcile the two pressures by planning in part, so that incremental pragmatics shapes the content and the first adjective, while the conventional order carries a communicative advantage that arises from the alternatives speakers produce.

By placing hierarchical composition and incremental pragmatics in one production model, we could test both accounts against the utterances that speakers produced, including which adjectives they selected and how many. The same model also captured redundant adjective use, which it derives from graded adjective meanings as in models of redundancy (Degen et al., 2020), so that adjective order and redundancy follow from one account of how speakers construct a useful referential utterance. Adjective order can therefore be understood only together with adjective selection, and the efficiency pressures on order act through the available alternatives that speakers could choose among. Together, these findings motivate an account of boundedly rational communication in which partial planning supports good-enough production. We discuss what this implies for each of the two accounts and for bounded planning in production, and finally turn to languages with postnominal adjectives.

6.1 Hierarchical Composition in Production

Scontras, Degen, and Goodman (2019, p. 7:19) asked whether subjectivity-based ordering preferences were conventionalised in the input over language evolution, remain active as speakers construct noun phrases, or arise from both. They themselves noted that studies of grounded adjectival modification could help to resolve this question. If the compositional pressure remains active, the noun-adjacent adjective should narrow the comparison class for size during production, so that *big* in *the big blue sticker* is evaluated among the blue objects. The present results favour the first of these options, in which the preferences were conventionalised over language evolution. Context updating did not improve the prediction of production, and its contribution to order was carried by the stable order term, so that the compositional pressure appears to act through the conventional order and need not be recomputed for each utterance. The simulation adds that such a convention need not rest on composition alone, as a size-first advantage for the listener also arose from the alternatives that speakers produce. Information-theoretic accounts likewise explain adjective order through properties of adjective and noun types, such as the mutual information between them (Hahn et al., 2018; Futrell, Dyer, and Scontras, 2020). As the noun was the same in every trial, such type-level pressures took the same value in every display, so in the present model they could act only through the stable order term.

A speaker could determine which objects count as big during scene apprehension and keep this comparison while selecting adjectives. Our task may have favoured such a fixed comparison, because the full display and the marked target were visible before production began, and the target could always be described as big. In discourse, a subset of referents may become relevant through a preceding utterance, potentially changing the context against which size is evaluated. Varying when contextual information becomes available would help identify when speakers retain an initial comparison and when they revise it. It is also not clear whether speakers and listeners use this interpretation strategy in online processing, and eye tracking could help to address this. The compositional benefit rests on uncertainty about what *big* means, which our discriminability manipulation approximated by varying how clearly the size difference could be seen. Independently varying lexical subjectivity and visual discriminability would further distinguish uncertainty about a property's interpretation from difficulty perceiving it in a particular context.

6.2 Incremental Pragmatics in Adjective Selection and Order

Incremental pragmatics proposes that an adjective is more useful when it helps the listener narrow down the referent as soon as it is produced (Fukumura, 2018; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). In the plan-guided model, this pressure acts through local competition at each position, which shaped which adjectives speakers mentioned and which came first. Fukumura (2018) proposed a similar competition, in which discriminability and availability raise the activation of a property, and more strongly activated properties are selected and

placed earlier (Fukumura, 2018, p. 47). That plan-guided production predicted better than fully incremental production suggests that incremental pragmatics alone does not account for these choices, as the value of the complete utterance also shapes which adjectives speakers produce and in which order. The model also underestimated how much high size discriminability increased size-initial responses. In the model, size discriminability matters only through how informative *big* is, whereas in Fukumura's account it also makes size more activated for the speaker, which can bring size to the first position directly.

When speakers mentioned both size and colour, colour-first ordering remained rare, as in earlier studies of size and colour (Danks and Schwenk, 1972). The size-first preference may be too conventionalised to be overridden by communicative constraints (Fukumura, 2018, p. 38), and the plan-guided model reflects this distinction through local competition and the stable order term. Redundant adjectives fit the same picture. Speakers added them more often when size alone identified the target under low size discriminability, as expected if a redundant adjective helps the listener (Rubio-Fernández, 2016; Degen et al., 2020), and the plan-guided model captured this pattern through graded adjective meanings. Rubio-Fernández (2016, p. 4) reported that redundant colour adjectives sped up listeners' identification of the referent in an eye-tracking study, as readily available information can guide visual search or hedge uncertainty about another property (Rubio-Fernandez, 2019; Tourtouri, Delogu, and Crocker, 2021).

6.3 Bounded Planning in Production

A useful utterance must be selected from alternatives that differ in content, length, and order. As the number of available properties increases, their combinations and orderings grow combinatorially, making exhaustive comparison increasingly demanding. The cognitive problem is how speakers preserve sensitivity to the communicative consequences of an utterance while limiting the alternatives and continuations they evaluate. Computational tractability constrains the processes and representations that can support this behaviour (Rooij, 2008); approximation alone provides no general guarantee of tractability (Kwisthout, Wareham, and Van Rooij, 2011). The same combinatorial demand arises for models of pragmatic reasoning. Rational Speech Act models are usually applied to small utterance spaces, such as two orders of the same adjectives, as each speaker probability requires interpreting every alternative and normalising over them. The present models push this approach much further. They interpret every prefix of fifteen utterances under recursive composition, nest a speaker that compares these prefixes inside a pragmatic listener in the simulation, and are estimated hierarchically across participants by Markov chain Monte Carlo, which repeats these computations at every step of the sampler. We made this feasible through an efficient implementation that computes all utterances and prefixes exactly in vectorised form, and by running inference on graphics processors with current machine-learning infrastructure (Bradbury et al., 2018; Phan, Pradhan, and Jankowiak, 2019).

Plan-guided production motivates a hypothesis about how planning could be organised under these constraints. An utterance-level plan could specify properties relevant to identifying the target, constraining the alternatives considered as each adjective is produced. The production of each adjective could remain sensitive to the developing utterance and the properties still available for mention. This combination of an utterance-level plan with adjective-by-adjective production accords with evidence that conceptual planning and linguistic formulation can proceed with different scopes (Griffin and Bock, 2000; Ferreira and Swets, 2002; Konopka and Meyer, 2014). It also connects to good-enough processing, in which representations can remain partial while serving the current task (Ferreira, Bailey, and Ferraro, 2002). Futrell (2023) proposed a model of incremental production in which the speaker combines a default policy, such as a language model, with communicative reward and a discounted value of future words. In this model, this discounting leads speakers to produce easily accessible words early. The plan-guided model has a similar structure, in which the stable order term acts as the default and informativeness as the communicative reward. Its parameter κ plays a role similar to the future discount, and the intermediate estimate is consistent with a speaker who discounts but does not ignore the value of later adjectives. The improvement from participant-specific κ came from two- and three-adjective utterances ($\Delta\mathrm{ELPD} = 229.4$, 95% CrI $[163.0, 319.3]$, and 183.8, $[100.8, 304.8]$), suggesting that the balance between evaluation of the complete utterance and local competition varies across speakers.

The present model also leaves aside influences that serve the speaker's own production. Speakers can rely on their own perspective when they formulate referring expressions (Horton and Keysar, 1996), and they may select and place early the properties that are most available or salient to themselves, which facilitates their own production (Fukumura, 2018, p. 38). The model represents such egocentric influences only through the salience term and the stable order term, so that part of what it attributes to informativeness could reflect availability. In addition, the data consist of completed descriptions, so the degree of planning describes the scope of evaluation that best predicts these descriptions and does not show directly how planning unfolds in time.

The model specifies a distribution over completed utterances and evaluates the full response space. A pro-

cess account must additionally explain how speakers generate candidate utterances and allocate computation to evaluating them. Manipulating planning time, and recording speech onset and eye movements, could test whether information in later adjectives affects which adjective speakers produce first.

6.4 Adjective Order in Postnominal Languages

The present data come from German, a prenominal language. In languages with postnominal adjectives, the noun-adjacent adjective is also the first adjective the listener hears. Subjectivity-based ordering preferences have been reported for the mirror-image order in such a language (Trainin and Shetreet, 2023). Incremental pragmatics, by contrast, still favours placing the more discriminating adjective first, as reported for Basque (Fukumura, 2018, p. 47). In a postnominal language, both accounts therefore favour placing colour directly after the noun when colour identifies the target, whereas they favour competing orders when size identifies the target, as composition places colour next to the noun and incremental pragmatics places size first. In a postnominal version of the simulation, in which composition follows the surface order, context updating lowered the size-first preference of global production with all fifteen utterances, from approximately 2.6 to 2.2 percentage points. Word order also affects which properties speakers mention. Rubio-Fernández, Mollica, and Jara-Ettinger (2021) found that English speakers produced more redundant colour adjectives than Spanish speakers, for whom colour follows the noun. Waldon and Degen (2021) captured this difference with an incremental RSA production model. Production data from postnominal languages could therefore test whether speakers resolve the competition in size-sufficient contexts mainly through which adjectives they select, as speakers in the present data did in colour-sufficient contexts. They could also test whether the size-first advantage that arose from the available alternatives in the simulation also holds when composition follows the surface order.

7 Data Availability Statement

Data, analysis scripts, model code, posterior summaries, and figure-generation scripts will be made available in an anonymised repository upon submission. The repository will include the production-model inputs and outputs, documentation of preprocessing and input reconstruction, and the scripts required to reproduce the reported behavioural and predictive analyses.

8 Author Note

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A Supplementary Experimental Methods and Results

A.1 Participants

All studies were conducted online via Prolific, with compensation set slightly above the platform's median level. Participants reported German as their native language.

Original slider study. The original slider data are those reported by Schlotterbeck and Wang (2023). The retained sample included 70 participants in the low- and 64 in the high-discriminability group. Exclusions addressed missing records, extreme completion times, and failures on control or task-relevance checks (Table 2).

Slider replication. The replication used the same task, conditions, and exclusion criteria, with newly generated items and an independent participant sample.

Production experiment. The retained production sample comprised 60 participants with low size discriminability and 53 with high size discriminability. Participants from either slider study were ineligible. Exclusions addressed extreme completion times, missing or unannotatable responses, and more than 80% identical responses.

Table 2 summarises the participant exclusions and the observations included in each analysis.

Table 2: Participant exclusions and observations included in each analysis.

Experiment	Starting and final N	Participant exclusions	Observations included in the manuscript
Original slider study	$150 \rightarrow 134$	More than three missing trial-list entries; extreme completion time; more than three control-item failures; more than three "none fits" responses on experimental items.	All retained critical trials: $10{,}606$ observations. Reported size–colour slider subset: $3{,}485$ observations.
Slider replication	$132 \rightarrow 104$	Same criteria as the original study.	All retained replication trials: $8{,}379$ critical observations. Reported size–colour slider subset: $2{,}786$ observations.
Production experiment	$149 \rightarrow 113$	Extreme completion time; more than 80% identical responses; more than 10% missing or unannotatable responses.	Full production dataset: $9{,}100$ annotated observations. Reported size–colour behavioural subset: $3{,}034$ observations.

A.2 Materials

The noun remained *Aufkleber*, and size, colour, and form varied across counterbalanced lists. Critical displays were realised through systematic permutations of colour and form values applied to 27 six-object display templates. This varied the surface displays while preserving the referential structure of each condition.

The slider stimulus inventory contains six lists of 180 main trials: 81 critical trials and 99 fillers and controls per list, plus three practice displays. Lists 1–3 use high size discriminability and lists 4–6 use low size discriminability. Each list contains nine critical trials for each combination of the three adjective families and three referential conditions. The large-size band comprises $\{9, 10\}$ in both groups; the smaller-size band is $\{1, 2, 3\}$ at high discriminability and $\{1, 2, 3, 4, 5, 6\}$ at low discriminability. Thus, contrasts between the large and smaller bands can span 6–9 recorded units in the high-discriminability materials and 3–9 units in the low-discriminability materials.

The slider-rating materials used the fixed adjective vocabulary in Table 3.

Table 3: Adjective vocabulary in the slider-rating item descriptions.

Property class	German adjective forms displayed in the item text
Size	*großen*, *kleinen*
Colour	*blauen*, *orangen*, *grauen*, *braunen*, *grünen*, *schwarzen*
Form	*runden*, *diamantförmigen*, *herzförmigen*, *quadratischen*, *sternförmigen*, *dreieckigen*

A.3 Design and Hypotheses

The studies shared the reference-game setting and display design described in Section 2. For slider judgements, greater referential usefulness of size predicts stronger size-first preference, yielding the ordering size-sufficient $>$ both-necessary $>$ colour-sufficient; a stable size-first preference predicts ratings above neutrality even when colour alone identifies the target. The slider interaction contrasts assess how discriminability effects differ across contexts, with the replication testing whether the pattern in the original data recurs.

A.4 Procedure

Each study began with three practice trials with feedback. Main trials were divided into four blocks of approximately equal length, with self-paced breaks and independently randomised trial order within each block. In the original slider study, participants completed 81 critical trials and 99 filler or control trials (180 total), with a mean completion time of 27.6 minutes. Control trials had an unambiguous expected response; the additional "none fits" option allowed participants to indicate that neither utterance identified the target. Frequent use of this option on experimental trials was an exclusion criterion, as detailed in Table 2. The replication used the same list and condition structure, with newly generated items.

On each critical trial, two phrases differing in adjective order appeared on opposite sides of a continuous slider, together with a "none of the descriptions fits" option. Left–right presentation was randomised across trials. Ratings were centred so that zero indicated neutrality, positive values favoured size-first, and negative values favoured colour-first order, on a scale from -0.5 to 0.5.

Participants typed adjective material into the frame *den* ___ *Aufkleber*. Practice trials introduced the relevant colour and form vocabulary. In the production experiment, participants completed 81 critical and 54 filler or control trials (135 total), with a mean completion time of 32.4 minutes.

Instructions. Participants in the production experiment read the following instructions, followed by a short reminder slide. The original German text is followed by our English translation.

> In diesem Experiment geht es darum, abgebildete Objekten zu beschreiben.
>
> Konkret geht es um das folgende Szenario: Sie öffnen mehrere Sets bestimmter Sammelaufkleber, und tauschen sich mit einer befreundeten Person am Telefon darüber aus. Die Aufkleber sehen unscheinbar aus, sind aber in bestimmten Kreisen extrem angesagt. Jedes Set enthält sechs Aufkleber. In Jedem Durchgang des folgenden Experiments werden Sie Ihren/Ihre Gesprächspartner/in fragen, ob er/sie einen bestimmten Aufkleber aus dem jeweiligen Set für seine/ihre eigene Sammlung braucht.
>
> Ihre Aufgabe im Experiment wird sein, in ein vorgegebenes Textfeld Beschreibungen von Aufklebern einzugeben, die Sie Ihr/e Gesprächspartner/in anbieten. Der Aufkleber, um den es jeweils geht, ist auf dem Bildschirm durch einen farbigen Rahmen hervorgehoben. Ihr/e Gesprächspartner/in kann zwar dieselben Aufkleber aus dem jeweiligen Set sehen. Aber weder sieht er/sie die farbige Markierung, noch ist ihm/ihr die räumliche Anordnung

Ihrer Aufkleber auf dem Bildschirm bekannt. Um sich auf einen bestimmten Aufkleber zu beziehen müssen Sie also dessen Aussehen beschreiben.

Wie das funktioniert sehen Sie gleich in einem kurzen Übungsblock.

Reminder slide: Denken Sie daran: Es geht darum, den hervorgehobenen Aufkleber im jeweiligen Aufkleberset anhand seines Aussehens zu beschreiben.

In this experiment, you will describe depicted objects.

Specifically, the scenario is as follows: You open several sets of particular collectible stickers and talk about them with a friend on the phone. The stickers look unremarkable but are extremely popular in certain circles. Each set contains six stickers. On each trial of the following experiment, you will ask your conversation partner whether they need a particular sticker from the current set for their own collection.

Your task in the experiment is to type into a text field descriptions of the stickers that you offer to your conversation partner. The sticker in question is highlighted on the screen by a coloured frame. Your conversation partner can see the same stickers from the current set. However, they see neither the coloured frame nor the spatial arrangement of your stickers on the screen. To refer to a particular sticker, you therefore have to describe its appearance.

You will see how this works in a short practice block.

Reminder slide: Remember: the task is to describe the highlighted sticker in each sticker set by its appearance.

A.5 Data Analysis

Response coding. Production responses were annotated using the 15 ordered adjective categories listed in Table 4. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size utterances, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table 4: Production annotation scheme. In the data files, `D` corresponds to the size adjective S; `C` and `F` correspond to colour and form.

Length	Codes
One adjective	`D`, `C`, `F`
Two adjectives	`DC`, `DF`, `CD`, `CF`, `FD`, `FC`
Three adjectives	`DCF`, `DFC`, `CDF`, `CFD`, `FDC`, `FCD`

The categorical response variable for the computational comparison is the ordered utterance label $y_i \in \{1, \ldots, 15\}$. For each binary behavioural outcome, matching responses were coded as 1 and all other annotated size–colour responses as 0.

Behavioural models. The behavioural hypotheses were evaluated with Bayesian hierarchical models. All behavioural analyses used the size–colour subsets: 3,485 observations in the original slider study, 2,786 in the replication, and 3,034 in the production experiment. For the slider analyses, ratings were reoriented toward size-first, divided by 100, and centred by subtracting 0.5. Let $\mathbf{x}_{ij}$ encode referential context, size discriminability, and their interaction, and let $\mathbf{z}_{ij}$ encode the participant intercept and context slopes. The slider models used a Gaussian likelihood with

$$y_{ij} \sim \mathrm{Normal}(\mu_{ij}, \sigma), \qquad \mu_{ij} = \mathbf{x}_{ij}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ij}^{\mathsf{T}}\mathbf{b}_i, \qquad \mathbf{b}_i \sim \mathrm{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathrm{participant}}).$$

For production, size-initiality indicated whether a coded response with more than one adjective began with the size adjective. Redundant adjective use indicated a three-adjective response in both-necessary contexts or a multi-adjective response in one-property-sufficient contexts. Both binary outcomes used a Bernoulli likelihood with a logit link and an item intercept c_k:

$$y_{ijk} \sim \mathrm{Bernoulli}(\pi_{ijk}), \qquad \mathrm{logit}(\pi_{ijk}) = \mathbf{x}_{ijk}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ijk}^{\mathsf{T}}\mathbf{b}_i + c_k, \qquad c_k \sim \mathrm{Normal}(0, \sigma_{\mathrm{item}}).$$

For the slider studies, the interaction was represented by two posterior contrasts: the high-minus-low discriminability effect in the both-necessary context minus the corresponding effect in the size-sufficient context, and the high-minus-low effect in the colour-sufficient context minus the effect in the size-sufficient context. The production interactions were planned contrasts on the posterior-predictive probability scale. For size-initiality, the contrast was the high-minus-low discriminability effect in the both-necessary context minus the mean effect in

the two one-property-sufficient contexts. For redundant adjective use, the contrast was the low-minus-high discriminability effect in the size-sufficient context minus the mean effect in the both-necessary and colour-sufficient contexts.

The slider models used a Gaussian likelihood, correlated participant intercepts and context slopes, fixed-effect priors $\mathrm{Normal}(0, .30)$, $\mathrm{HalfNormal}(.20)$ priors on participant standard deviations, and a $\mathrm{HalfNormal}(.25)$ prior on residual standard deviation. The production models used Bernoulli likelihoods, the same correlated participant effects, an item intercept, a $\mathrm{Normal}(0, 2)$ intercept prior, $\mathrm{Normal}(0, 1.5)$ priors on the remaining fixed effects, and $\mathrm{HalfNormal}(1)$ priors on participant and item standard deviations. Participant-effect correlations received an $\mathrm{LKJ}(2)$ prior. Four chains run in parallel each provided $1{,}000$ retained draws after $1{,}000$ warm-up iterations. All four fits had zero divergences, no maximum-tree-depth hits, $\widehat{R} \leq 1.01$, and minimum effective sample sizes above 900. For each posterior draw, posterior-predictive cell expectations were computed for a new participant and, in production, a new item by integrating over the fitted random-effects distributions. The planned contrasts were then evaluated on the rating or response-probability scale.

A.6 Detailed behavioural results

Tables 5 and 6 report posterior medians and 95% credible intervals for the behavioural contrasts. Slider estimates are in centred-rating units; production estimates are differences in response probability. The column P reports the posterior probability of the direction stated in each row; a dash indicates that a directional probability was not separately reported for that contrast in the main analysis summary. Interaction estimates compare effects across contexts, whereas within-context contrasts compare the two discriminability groups in one context.

A.6.1 Slider ratings and replication

Both slider studies supported the predicted referential-context gradient, including a residual size-first preference in colour-sufficient displays (Figure 8). The original data also showed a credible interaction in the comparison of size-sufficient and both-necessary displays. Higher discriminability was associated with a shift towards size-first ratings in size-sufficient displays and towards colour-first ratings in both-necessary displays, although the credible intervals for both within-context effects included zero. The interaction therefore concerned the difference between these effects; each individual effect was less precisely estimated. The corresponding interaction between colour-sufficient and size-sufficient displays was less conclusive.

The replication did not clearly reproduce either interaction contrast. Its estimated effect of higher size discriminability was positive in all three contexts, with intervals above zero in size-sufficient and colour-sufficient displays and an interval spanning zero in both-necessary displays. Thus, clearer size differences could strengthen size-first preference in this replication, with uncertainty about variation in that effect across contexts. The two studies used the same Bayesian hierarchical model specification. Estimates in Table 5 are posterior medians, with contrasts expressed in centred-rating units; directional probabilities refer to the stated contrast under the fitted model.

The posterior probability of the graded context order was greater than .999 in the original study and .999 in the replication. Observed mean centred ratings in the replication were .272 in size-sufficient, .197 in both-necessary, and .054 in colour-sufficient contexts. Figure 9 shows the observed ratings in the original study; Figure 8 shows fitted cell expectations with 95% credible intervals for both slider studies.

A.6.2 Production contrasts

The planned redundancy interaction showed a larger increase under low discriminability in size-sufficient displays than in the other two contexts (Figure 3, Panel B; Table 6). Within size-sufficient displays, lower discriminability increased redundant adjective use; the corresponding increases in both-necessary and colour-sufficient displays were smaller, and their credible intervals included zero.

For size-initiality, the high-minus-low discriminability effect was larger when both properties were necessary than when one property was sufficient (Figure 3, Panel C). Higher discriminability increased size-initial responses in both-necessary displays. The effects in size-sufficient and colour-sufficient displays were uncertain. Averaged across discriminability conditions, size-initial responses were more probable in size-sufficient than in both-necessary displays, and more probable in both-necessary than in colour-sufficient displays. Among responses containing both size and colour, size preceded colour in most responses in every condition.

Observed redundancy proportions in size-sufficient displays were .90 under low and .63 under high size discriminability. Observed size-initial proportions in both-necessary displays were .56 under low and .70 under high discriminability. These descriptive proportions differ from the posterior-predictive contrasts in Table 6, which integrate over participant and item variation.

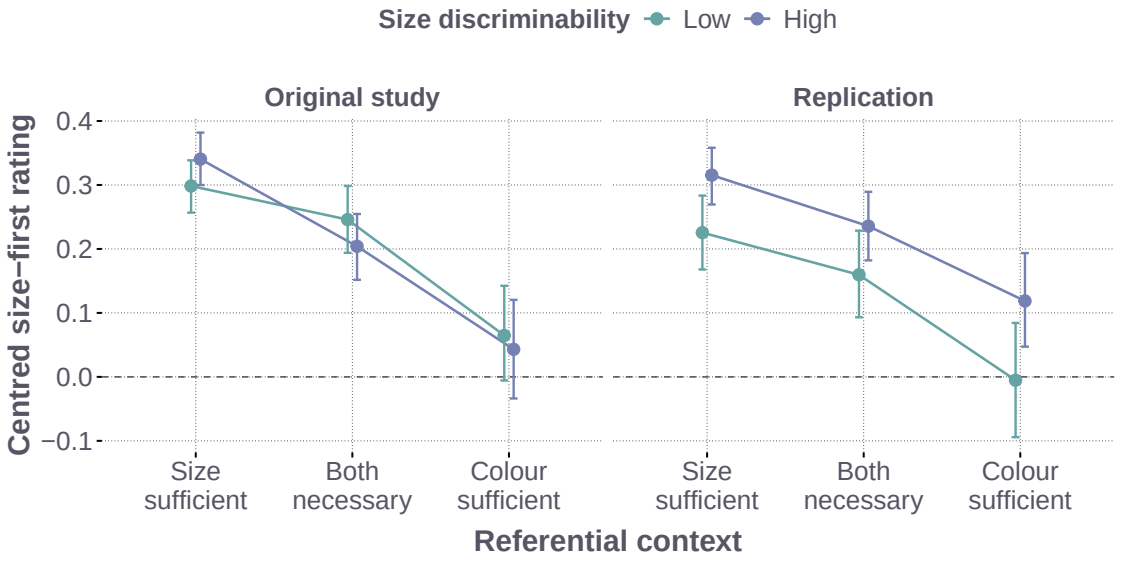


Figure 8: Ordering judgements in the original slider study and its replication by referential context and size discriminability. Points show posterior medians of expected centred ratings for a new participant; error bars show 95% credible intervals. Positive values indicate size-first preference; zero indicates neutrality. The two studies use the same model specification and interval definition.

Table 5: Slider-rating contrasts, with interactions followed by context contrasts and within-context discriminability effects. Discriminability effects are high minus low. Interaction rows subtract the size-sufficient discriminability effect from the effect in the named context.

Study	Contrast	Median	95% CrI	P (direction)
Original	Interaction: both-necessary	$-.085$	$[-.151, -.022]$	.995 (< 0)
Original	Interaction: colour-sufficient	$-.063$	$[-.172, .039]$	.885 (< 0)
Original	Size-sufficient minus both-necessary	.094	$[.061, .127]$	—
Original	Both-necessary minus colour-sufficient	.171	$[.131, .208]$	—
Original	Colour-sufficient rating relative to zero	.054	$[.003, .109]$	.982 (> 0)
Original	Discriminability: size-sufficient	.042	$[-.013, .099]$	—
Original	Discriminability: both-necessary	$-.043$	$[-.118, .028]$	—
Original	Discriminability: colour-sufficient	$-.021$	$[-.129, .082]$	—
Replication	Interaction: both-necessary	$-.014$	$[-.097, .071]$	.623 (< 0)
Replication	Interaction: colour-sufficient	.034	$[-.082, .157]$	.285 (< 0)
Replication	Size-sufficient minus both-necessary	.073	$[.029, .115]$	—
Replication	Both-necessary minus colour-sufficient	.140	$[.090, .190]$	—
Replication	Colour-sufficient rating relative to zero	.057	$[.001, .116]$	.977 (> 0)
Replication	Discriminability: size-sufficient	.090	$[.014, .160]$	—
Replication	Discriminability: both-necessary	.076	$[-.011, .161]$	—
Replication	Discriminability: colour-sufficient	.124	$[.016, .240]$	—

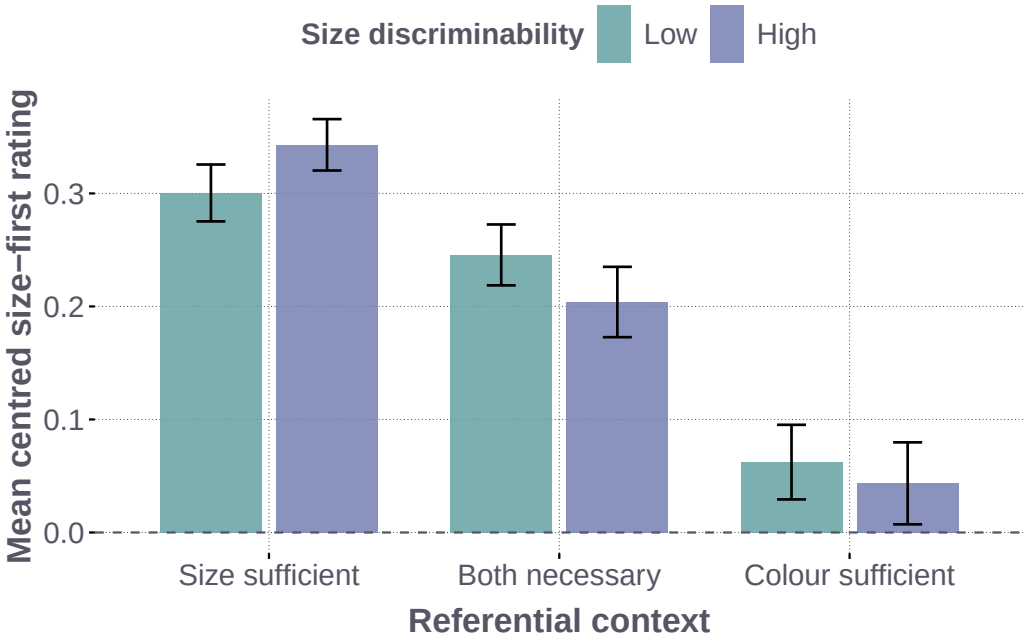


Figure 9: Observed ratings in the original slider study. Mean centred trial ratings by referential context and size discriminability, with descriptive 95% t intervals; positive values indicate size-first preference.

Table 6: Production contrasts on the response-probability scale. The redundancy interaction compares the low-minus-high difference in size-sufficient displays with its mean in the other two contexts. The size-initiality interaction compares the high-minus-low difference in both-necessary displays with its mean in the two one-property-sufficient contexts.

Contrast	Median	95% CrI	P (direction)
Redundancy: planned interaction	.153	[.072, .236]	$>$ .999 ($>$ 0)
Redundancy: size-sufficient, low minus high	.234	[.151, .318]	$>$ .999 ($>$ 0)
Size-initiality: planned interaction	.159	[.067, .251]	$>$ .999 ($>$ 0)
Size-initiality: both-necessary, high minus low	.144	[.043, .244]	.997 ($>$ 0)
Size-initiality: size-sufficient, high minus low	.043	[−.045, .134]	—
Size-initiality: colour-sufficient, high minus low	−.072	[−.171, .032]	—
Size-initiality: size-sufficient minus both-necessary	.180	[.126, .234]	—
Size-initiality: both-necessary minus colour-sufficient	.479	[.396, .556]	—

A descriptive check of the complete production data also showed group differences in the colour–form family. In form-sufficient displays, form-initial responses were more frequent in the high- than in the low-discriminability group (.77 vs. .71). These differences can also reflect group variation in how the task was approached, as discriminability was manipulated between participants. The production model represents participant variation alongside the visual manipulation.

B Production Model Specification

B.1 Representation of Experimental Displays

The model represents each recorded display as a 6×3 array of size, colour, and form values, with the intended referent indexed first.

B.2 Graded Size Meaning

Size is represented by a graded threshold over the comparison class C_t (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019):

$$\theta_\delta(C_t) = Q_{.80}(C_t) - \delta\left(Q_{.80}(C_t) - Q_{.20}(C_t)\right), \tag{7}$$

where $Q_{.20}$ and $Q_{.80}$ are probability-weighted size quantiles and δ controls threshold selectivity. For an object of size s_o, graded membership is

$$m_{\mathrm{S}}(o \mid C_t) = \Phi\left(\frac{s_o - \theta_\delta(C_t)}{w\sqrt{s_o^2 + \theta_\delta(C_t)^2 + \epsilon}}\right), \qquad w = .6856, \quad \epsilon = 10^{-8}, \tag{8}$$

where Φ is the standard normal cumulative distribution function. The weighted quantiles are the first observed sizes at which cumulative comparison-class probability reaches .20 and .80. The same width w is used in both discriminability groups; the recorded sizes enter this function directly. For colour and form, matching objects receive membership ν and nonmatching objects $1 - \nu$, with $\nu_{\mathrm{C}} = .59$ and $\nu_{\mathrm{F}} = .50$. Each prefix is interpreted from the uniform object prior, applying its modifiers once in noun-outward order. Context updating changes the comparison-class weights used for the size quantiles as inner modifiers are applied. Equations (1), (2) and (5) define how this meaning enters the literal interpretation L and the three production accounts.

B.3 Component Inventory

Table 7: Components of the production models.

Component	Status	Role
Observed display input	Fixed	All visible size, colour, and form values
Size threshold and blur	Fixed constants	Graded size meaning in the display comparison class
Colour reliability	Fixed	Graded support for matching colour
Form reliability	Fixed, relaxed	Form production supported by the length and selection terms
Pragmatic rationality	Estimated, hierarchical	Participant variation in response concentration
Stable-order score Ω	Fixed feature	Relative order expectation within adjective sets
Stable-order weight β_i	Estimated, hierarchical	Participant variation in stable order expectation
Salience coefficient	Estimated	Display prominence in candidate utility
Length and selection coefficients	Estimated	Support for sufficient single adjectives, form use, and length and discriminability effects
κ or κ_i	Estimated in plan-guided production	Weight of local adjective competition, shared or participant-specific

The form semantics, stable-order feature, and length and selection terms were specified on the present data and are identical across all compared production architectures and semantic regimes. The population pragmatic-optimality parameter has prior $\alpha_0 \sim \mathrm{HalfNormal}(3)$, and the population log stable-order weight has prior $\mu_{\log\beta} \sim \mathrm{Normal}(0, .35)$. Participant effects are noncentred: $\alpha_i = \alpha_0 \exp(\tau_{\log\alpha} z_{\alpha i})$ and $\beta_i = \exp(\mu_{\log\beta} + \tau_\beta z_{\beta i})$, with independent standard-normal z values, $\tau_{\log\alpha} \sim \mathrm{HalfNormal}(.15)$, and $\tau_\beta \sim \mathrm{HalfNormal}(.5)$. The shared plan-guided weight has prior $\kappa \sim \mathrm{Uniform}(0, 1)$. In the model with participant-specific κ_i, $\mu_\kappa \sim \mathrm{Uniform}(0, 1)$ and $\kappa_i = \mathrm{logit}^{-1}(\mathrm{logit}(\mu_\kappa) + \tau_\kappa z_{\kappa i})$, with $\tau_\kappa \sim \mathrm{HalfNormal}(.5)$ and independent $z_{\kappa i} \sim \mathrm{Normal}(0, 1)$. The salience coefficient has a $\mathrm{HalfNormal}(1)$ prior and the salience-continuation coefficient a $\mathrm{HalfNormal}(.75)$ prior. The sufficient-single-adjective and form-use coefficients each have a $\mathrm{HalfNormal}(1.5)$ prior; the three-adjective penalty and the two size-reliability coefficients each have a $\mathrm{HalfNormal}(1)$ prior. Threshold selectivity is fixed at $\delta = .50$; colour reliability, form reliability, and perceptual blur take the values specified above.

Visual salience. The salience feature combines a fixed preference across properties with their contrast in the display. For the target o^*, let s_o, c_o, and f_o be an object's recorded size, colour, and form, and let h indicate high size discriminability. The three contrast scores are

$$r_{\mathrm{S}} = \frac{1+h}{2}\ \mathrm{logistic}\left(\frac{s_{o^*} - \max_{o\neq o^*} s_o}{\mathrm{sd}(s) + 10^{-8}}\right), \tag{9}$$

$$r_{\mathrm{C}} = 1 - \frac{1}{5}\sum_{o\neq o^*} \mathbf{1}\{c_o = c_{o^*}\}, \qquad r_{\mathrm{F}} = 1 - \frac{1}{5}\sum_{o\neq o^*} \mathbf{1}\{f_o = f_{o^*}\}. \tag{10}$$

Here $\mathrm{sd}(s)$ is the population standard deviation across the six displayed sizes. The fixed baseline scores are $(b_{\mathrm{S}}, b_{\mathrm{C}}, b_{\mathrm{F}}) = (0, 1, .25)$. Centring across the three properties gives

$$z_a = b_a + r_a - \frac{1}{3}\sum_v (b_v + r_v).$$

A property receives greater salience when it contrasts with the distractors; the size contrast is additionally scaled by discriminability. The same feature enters prefix utility through $\lambda_{\mathrm{salience}} z_a$ and the continuation cost in the length and selection terms below.

Stable order expectations. We derived the fixed order feature from `dbmdz/german-gpt2`. For each pairing of six colour and six form adjectives, the scoring script generated all fifteen non-empty orders using those adjectives and *großer*, followed by the noun *Aufkleber*, without a surrounding sentence. It evaluated the resulting 540 strings, including repeated shorter strings, using mean token surprisal in bits. We averaged this surprisal within each order category and normalised $2^{-\bar{s}(u)}$ across the fifteen categories to obtain the fixed weights $q(u)$. For $\mathcal{U}_{A(u)}$, the set of orders expressing the same adjective set as u, the model uses

$$\Omega(u) = \log q(u) - \frac{1}{|\mathcal{U}_{A(u)}|}\sum_{v\in\mathcal{U}_{A(u)}} \log q(v). \tag{11}$$

This centring preserves relative order preferences within an adjective set and gives every single-adjective response a score of zero. The scoring script, archived token-surprisal table, aggregation code, and fixed scores and residuals are supplied with the model code. Participant-specific β_i determines the influence of these stable expectations when the model normalises across the complete response space.

Length and selection terms. The length and selection terms $\Lambda(u, C)$ connect the referential design to choices about which properties to mention and how far to continue an utterance. Let $T = |u|$ and $A(u)$ denote the length and unordered adjective set of u. For this term, C includes the display's condition labels: $d \in \{\mathrm{S}, \mathrm{C}, \mathrm{F}, \varnothing\}$ is the property that identifies the target on its own, with $d = \varnothing$ in both-necessary conditions; c indicates a colour-sufficient condition; and $h = 1$ indicates high size discriminability ($h = 0$ for low). Using $\mathbf{1}\{\cdot\}$ for an indicator, the production models use

$$\begin{aligned}\Lambda(u, C) = \ & -\rho_{\mathrm{cont}} \sum_{t=1}^{T-1} \max(z_{w_t}, 0) + \lambda_{\mathrm{suff}} \mathbf{1}\{d \neq \varnothing,\ u = (d)\} \\ & + \lambda_{\mathrm{F}}(1-c)\mathbf{1}\{\mathrm{F} \in A(u)\} - \lambda_3 \mathbf{1}\{T = 3\} \\ & + \lambda_{\mathrm{S}} h\, \mathbf{1}\{d = \mathrm{S},\ u = (\mathrm{S})\} \\ & + \lambda_{\mathrm{SF}}(1-2h)\mathbf{1}\{d = \mathrm{S},\ A(u) = \{\mathrm{S}, \mathrm{F}\}\}.\end{aligned} \tag{12}$$

The continuation cost sums the positive salience scores of adjectives followed by another adjective, using the same z scores as the prefix utility in Equation (2). The sufficiency term favours a single adjective that identifies the target, while the form term increases the value of form-containing utterances outside colour-sufficient conditions. The three-adjective penalty offsets the value accumulated along longer utterances. In size-sufficient conditions, the final two terms additionally favour the single size adjective at high discriminability and favour size–form pairs at low discriminability while penalising them at high discriminability. The pair term applies equally to both orders. All six coefficients are nonnegative and shared across participants within each fitted model, with the priors given above. These terms enter the completed-utterance score alongside accumulated prefix utility, local competition, and the stable-order score; normalisation over all fifteen responses turns their combined values into predictions of adjective selection, length, and order.

Response likelihood. Writing $g(u)$ for accumulated prefix utility, $i(u)$ for the sum of local log probabilities, and $B(u) = \beta_i \Omega(u) + \Lambda(u, C)$, the computed score before Laplace smoothing is $(1 - \kappa_i)g(u) + \kappa_i i(u) + B(u)$. Normalising these scores yields $\widetilde{S}_{\kappa_i}(u \mid C)$, whose endpoints obey the geometric interpolation in Equation (5). A final Laplace smoothing step mixes this distribution with a uniform distribution over the fifteen responses, giving the categorical response likelihood

$$p(y_{in} = u \mid C_{in}) = (1 - .003)\widetilde{S}_{\kappa_i}(u \mid C_{in}) + \frac{.003}{15}. \tag{13}$$

The matched global and fully incremental models fix κ at zero and one, respectively; the matched plan-guided model estimates a shared κ; the model with participant-specific κ_i estimates them for each participant. All models retain participant hierarchies for pragmatic optimality and stable-order weight, with the remaining task coefficients shared across participants.

C Production Model Comparison and Diagnostics

C.1 Inference and Predictive Comparison

Each likelihood evaluation computes all fifteen completed utterances exactly, with the listener and speaker computations vectorised and compiled in JAX. All eight production fits used four No-U-Turn Sampler (NUTS) chains on a GPU and 1,000 warm-up iterations per chain. The two global models retained 4,000 draws per chain at maximum tree depth 8; the other six models retained 1,000 draws per chain at depth 6. The eight fits had zero divergences, maximum $\widehat{R} < 1.01$, minimum bulk and tail effective sample sizes above 400, minimum Bayesian fraction of missing information (BFMI) above .64, and no observation with Pareto $k > .7$.

The pointwise leave-one-out score vectors use the same trial ordering and response support. Because observations are repeated within participants, paired score contributions are first summed within participant. A participant-level Bayesian bootstrap then assigns Dirichlet weights to the 113 participant contributions over 200,000 draws. Reported uncertainty comprises 95% credible intervals and posterior probabilities for directional contrasts. The predictive target is another response from a participant represented in the dataset, with that participant's remaining trials available to estimate their participant-specific parameters.

Table 8: Bayesian-bootstrap contrasts for the matched production architectures and the plan-guided participant hierarchy. Interaction contrasts compare the updating-minus-fixed contrast between two production architectures.

Contrast	ΔELPD	95% credible interval	$P(\Delta > 0)$
Plan-guided, shared κ, minus average endpoints	402.677	[311.466, 503.098]	$> .999$
Fully incremental minus global	−291.596	[−684.673, −1.321]	.024
Updating minus fixed, shared-κ grid	−43.527	[−56.005, −31.770]	$< .001$
Participant κ_i hierarchy, context-fixed	364.742	[248.178, 520.967]	$> .999$
Participant κ_i hierarchy, updating	372.285	[255.597, 526.056]	$> .999$
Simple effects and interaction, shared κ			
Plan-guided minus global, context-fixed	265.168	[77.902, 417.166]	.994
Plan-guided minus fully incremental, context-fixed	567.420	[364.061, 815.282]	$> .999$
Plan-guided minus global, updating	248.589	[69.599, 393.457]	.994
Plan-guided minus fully incremental, updating	529.530	[332.673, 772.631]	$> .999$
Updating minus fixed, global	−45.104	[−60.182, −30.157]	$< .001$
Updating minus fixed, plan-guided	−61.683	[−80.920, −43.573]	$< .001$
Updating minus fixed, fully incremental	−23.793	[−32.383, −16.102]	$< .001$
Interaction, plan-guided minus global	−16.579	[−25.733, −6.678]	.001
Interaction, plan-guided minus fully incremental	−37.890	[−54.992, −21.627]	$< .001$
Interaction, global minus fully incremental	−21.311	[−36.263, −6.518]	.003

Averaging over the two semantic regimes, the plan-guided model had the highest predictive accuracy of the three architectures in .994 of the Bayesian-bootstrap draws. In the context-fixed plan-guided model, the posterior mean of the shared κ was

$$\kappa = .446, \qquad 95\% \text{ credible interval} = [.407, .486]. \tag{14}$$

C.2 Posterior Parameter Estimates

Table 9 reports the population-level posteriors of the best-predicting model, and Table 10 reports the posterior means of the same parameters in the six matched models.

Table 9: Posterior estimates of the population-level parameters in the best-predicting model, plan-guided production with context-fixed semantics and participant-specific κ_i.

Parameter	Description	Mean	95% credible interval
α_0	Population pragmatic optimality	.924	$[.769, 1.093]$
$\tau_{\log \alpha}$	Participant scale of $\log \alpha_i$	.800	$[.688, .922]$
$\mu_{\log \beta}$	Population log stable-order weight	.962	$[.888, 1.037]$
τ_β	Participant scale of $\log \beta_i$	.353	$[.297, .420]$
μ_κ	Population degree of planning	.472	$[.337, .618]$
τ_κ	Participant scale of logit κ_i	2.720	$[2.277, 3.207]$
$\lambda_{\text{salience}}$	Salience	2.075	$[2.006, 2.145]$
ρ_{cont}	Continuation cost	.003	$[.000, .010]$
λ_{suff}	Sufficient single adjective	1.817	$[1.729, 1.907]$
λ_{F}	Form use outside colour-sufficient contexts	2.537	$[2.437, 2.634]$
λ_3	Three-adjective penalty	.605	$[.530, .677]$
λ_{S}	Single size adjective, high discriminability	1.596	$[1.385, 1.805]$
λ_{SF}	Size–form pair by discriminability	.080	$[.004, .189]$

Table 10: Posterior means of the population-level parameters in the six matched models. The global and fully incremental models fix κ at zero and one.

	Context-fixed			Sequential context updating		
Parameter	Global	Plan-guided	Fully incr.	Global	Plan-guided	Fully incr.
α_0	.525	.693	.961	.495	.630	.857
$\tau_{\log \alpha}$	.969	1.034	.523	.971	1.035	.552
$\mu_{\log \beta}$	.839	.911	.857	.818	.893	.865
τ_β	.370	.334	.332	.364	.331	.328
κ	–	.446	–	–	.431	–
$\lambda_{\text{salience}}$	2.181	1.975	1.464	2.133	1.949	1.463
ρ_{cont}	.001	.002	.004	.001	.002	.005
λ_{suff}	1.872	1.775	2.232	1.896	1.814	2.238
λ_{F}	2.657	2.520	2.031	2.624	2.480	2.021
λ_3	.551	.551	1.142	.607	.609	1.135
λ_{S}	1.674	1.484	.973	1.735	1.559	.988
λ_{SF}	.135	.110	.081	.107	.087	.078

C.3 Decomposing the Semantic Predictive Contrast

The architecture-specific updating-minus-fixed effects were -45.104 ELPD for global production, -61.683 for plan-guided production, and -23.793 for fully incremental production. Each architecture-specific 95% credible interval excluded zero. The sequentially updating participant-specific model was lower by 54.140 ELPD, with a 95% credible interval for updating minus fixed of $[-69.042, -39.733]$.

Let $\ell(u)$ denote utterance length and $A(u)$ its unordered adjective set. For each held-out response, its predictive probability factors as

$$p_{-in}(u \mid C_{in}) = p_{-in}(\ell(u) \mid C_{in})\, p_{-in}(A(u) \mid \ell(u), C_{in})\, p_{-in}(u \mid A(u), C_{in}). \tag{15}$$

We first integrate the fifteen-response distribution using the PSIS weights for leaving out that entire response, then obtain the length and adjective-set probabilities by summation. Taking logarithms yields three additive contributions to the original response score. Paired differences are summed within participant, with 95% intervals from 200,000 Dirichlet-bootstrap draws. For the participant-specific κ_i pair, updating minus fixed was -21.549 ELPD for length, 95% credible interval $[-30.933, -12.443]$, and -31.068 for adjective set conditional on length, $[-39.202, -23.530]$. The conditional-order contribution was -1.524 ELPD, with an interval of $[-4.879, 1.774]$. Across all four semantic pairs, adjective set conditional on length makes a negative contribution to updating minus fixed. For the same pair, the original response contrast sums to -17.548 ELPD in size–colour trials, -51.172 in size–form trials, and 14.580 in colour–form trials.

Response residuals. For the two plan-guided models with participant-specific κ_i, mean absolute residuals across response cells were 3.04 percentage points under context-fixed semantics and 3.08 under updating. The largest between-model differences occurred in size-sufficient displays, in line with the held-out partition, which located the largest net disadvantage of updating in the size–form family.

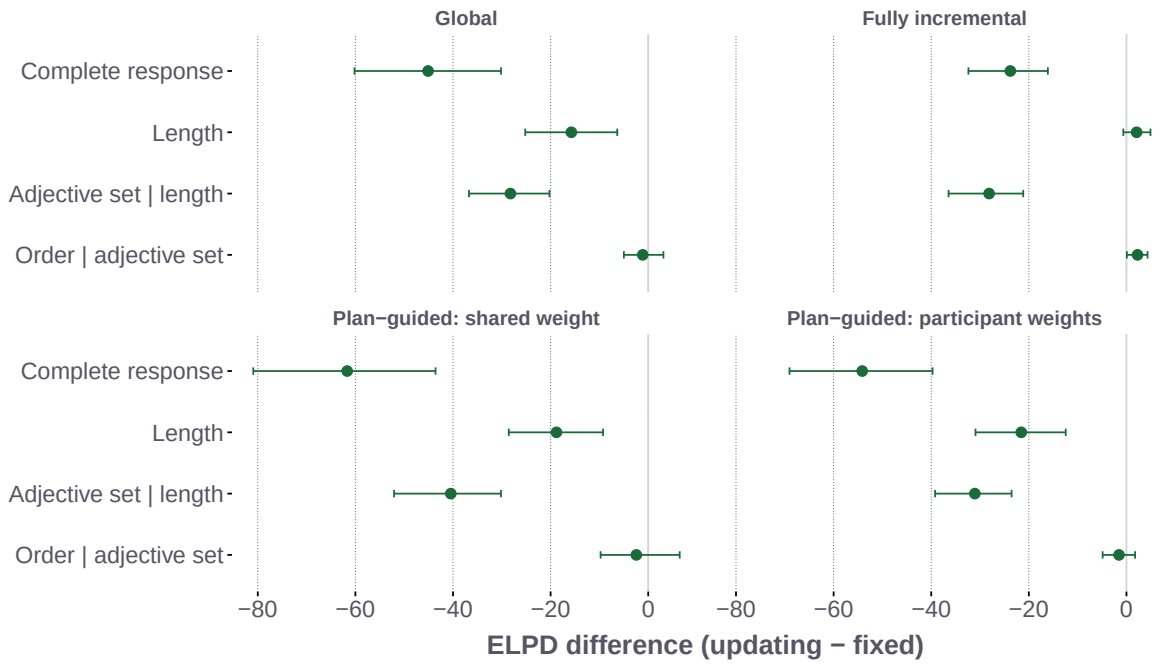


Figure 10: Additive decomposition of updating-minus-fixed response ELPD for each matched semantic pair. Length, adjective set conditional on length, and order conditional on the adjective set sum to the complete-response contrast. Points show paired score differences and intervals show participant-level Bayesian-bootstrap 95% credible intervals. The final panel uses participant-specific κ_i; the other plan-guided panel uses a shared κ.

C.4 Posterior-Predictive Adequacy

The condition-wise response distributions appear in Figure 11. Across the 270 cells formed by the fifteen completed utterances, three adjective families, three referential contexts, and two size-discriminability conditions, the posterior predictions correlated with the observed proportions at $r = .909$, with a mean absolute error of 3.0 percentage points and a root mean squared error of 6.0 percentage points.

In size–colour trials, the model reproduced the predominance of colour-only responses when colour was sufficient and the greater use of size alone when size was sufficient and highly discriminable. The main remaining discrepancy concerned the balance between some two- and three-adjective utterances, with size–form responses overpredicted and size–colour–form responses underpredicted.

The same comparison for size–form and colour–form trials appears in Figure 12.

C.5 Robustness Checks

Seven robustness checks address the participant-level structure of the architecture comparison, its distribution across adjective families, the fixed colour reliability, the condition-specific length and selection terms, the prefix utility, the stable order term, and generalisation to new participants. Unless stated otherwise, contrasts are context-fixed, summed within participant, and summarised with the participant-level Bayesian bootstrap described above. In total, the comparisons rest on 72 fitted production models: the eight models in Section 4.1, 50 cross-validation fits, and 14 refits for the robustness checks below.

Endpoint mixture benchmark. For each participant, the benchmark takes the higher held-out score of the global and the fully incremental model. Because this choice is made after seeing the scores, the benchmark favours an account in which participants differ only in which endpoint they resemble. The plan-guided model with participant-specific κ_i predicted better than this benchmark, whereas the plan-guided model with a shared κ did not (Table 11). The participant-specific model exceeded both endpoint models for 59 of the 113 participants.

Adjective families. The plan-guided advantage over the average of the two endpoint models was credible in each adjective family. Within size–colour trials, it was larger for responses that contained a form adjective than for responses that did not, but it was credible in both.

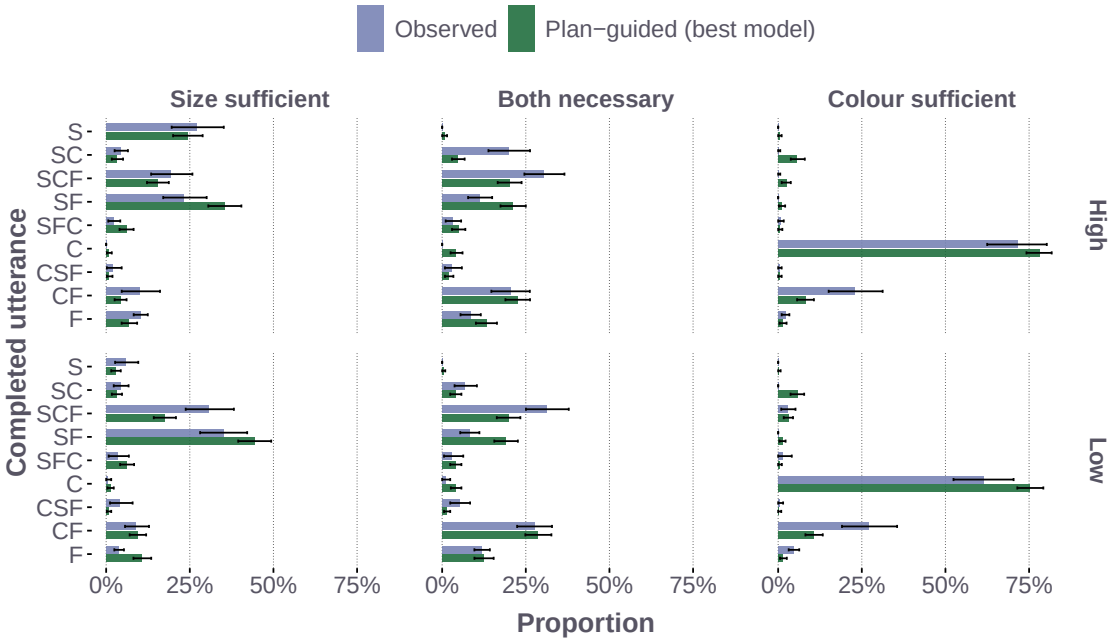


Figure 11: Observed and posterior-predicted proportions for frequent utterances in size–colour trials under the best-predicting plan-guided model. The figure includes utterances whose observed or mean posterior-predicted proportion reaches 5% in at least one context-by-discriminability condition, with proportions pooled across trials within each condition. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Adjective labels preserve surface order: S denotes size, C colour, and F form.

Colour reliability. The plan-guided models with context-fixed and updating semantics were refitted with colour reliability $\nu_{\mathrm{C}} = .75$ instead of .59. Both models predicted worse than at $\nu_{\mathrm{C}} = .59$ ($\Delta\mathrm{ELPD} = -263.0$, 95% CrI $[-365.7, -163.4]$ for context-fixed semantics), and updating still predicted worse than context-fixed semantics. The shared κ remained between the endpoints (posterior mean .29).

Condition-specific terms. The global, plan-guided, and fully incremental models were refitted under both semantic regimes with the two condition-specific length and selection terms fixed at zero. These models predicted worse than their counterparts with the two terms, but plan-guided production still predicted better than both endpoints and context-fixed semantics still predicted better than updating. The shared κ remained between the endpoints (posterior mean .50, 95% CrI $[.46, .55]$ under context-fixed semantics).

Prefix utility. The global and plan-guided models were refitted under both semantic regimes with a terminal utility, which scores a completed utterance only by its final target probability, while salience and local competition entered as before (Appendix G.1). With the terminal utility, the global model predicted considerably worse than with the prefix utility, whereas the plan-guided model predicted about as well (Table 11). The shared κ rose to .80 (95% CrI $[.75, .86]$), so that local competition took over the information that the prefix utility otherwise adds at each adjective position. Plan-guided production still predicted better than global production, and context-fixed semantics still predicted better than updating.

Stable order term. The plan-guided models were refitted under both semantic regimes with the stable-order weight β fixed at zero for all participants. Without this term, both models predicted far worse than their counterparts with it, but updating predicted better than context-fixed semantics. The shared κ moved close to the global endpoint under both semantic regimes (posterior mean .004).

New participants. Participants were assigned to ten folds, stratified by size-discriminability group. Each model was refitted without the participants of one fold, and every held-out participant was scored as a new speaker. The score integrates over participant effects drawn from the fitted population distribution and is computed jointly over all of that participant's responses, using 400 posterior draws and 100 participant-effect draws per posterior draw.

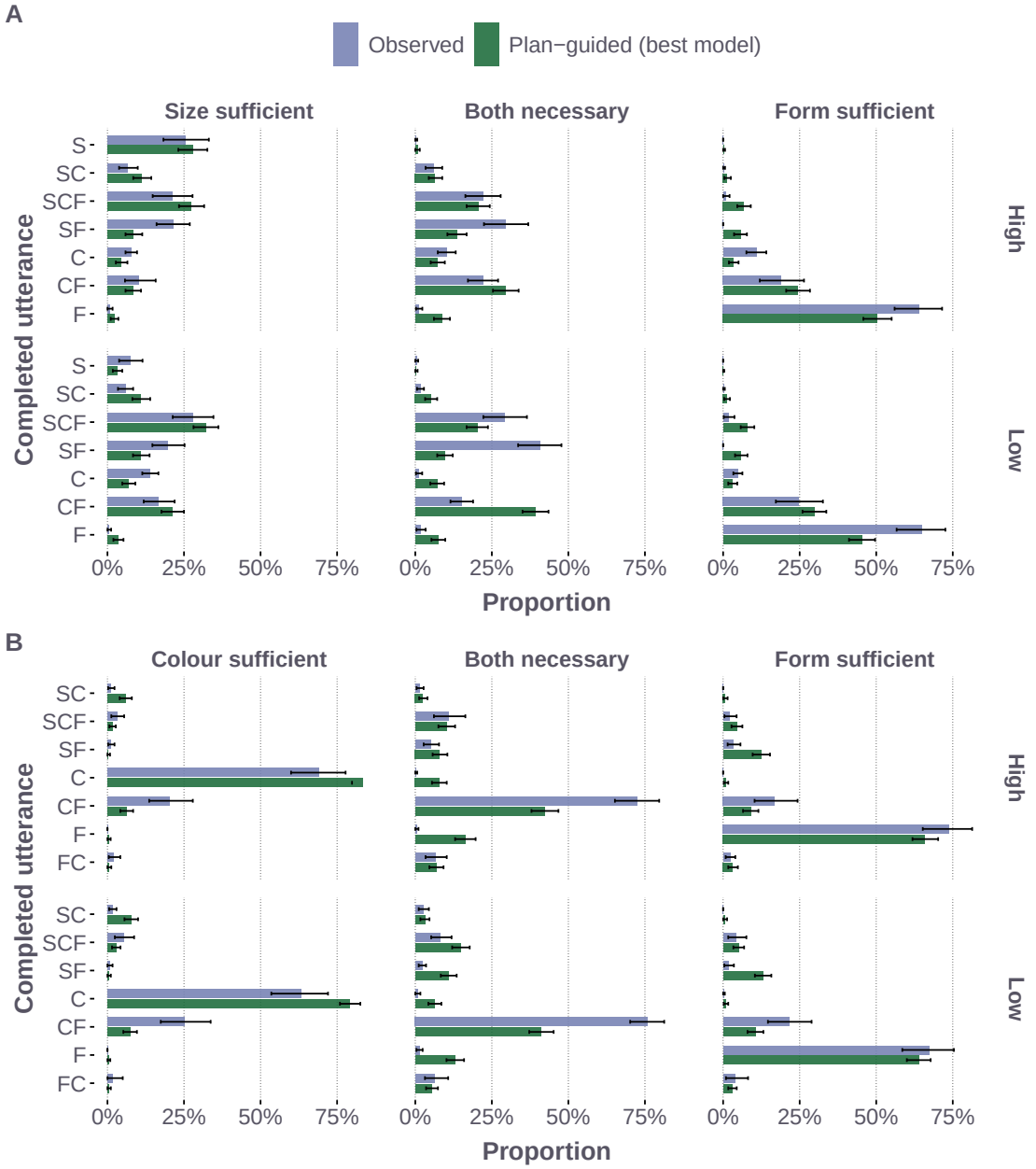


Figure 12: Observed and posterior-predicted proportions for frequent utterances in size–form (Panel A) and colour–form trials (Panel B) under the best-predicting plan-guided model. Utterance selection, pooling, and intervals follow Figure 11. Adjective labels preserve surface order: S denotes size, C colour, and F form.

All 50 fits had no divergences and a maximum $\widehat{R}$ of 1.011; halving the number of participant-effect draws changed fold totals by less than 0.5 ELPD.

Table 11: Robustness checks for the production model comparison. Differences are in ELPD, summed over participants, with 95% credible intervals from the participant-level Bayesian bootstrap. Rows use context-fixed semantics unless stated otherwise.

Contrast	ΔELPD	95% CrI
Endpoint mixture benchmark		
Participant-specific κ_i minus benchmark	249.7	$[103.8, 416.1]$
Shared κ minus benchmark	-115.0	$[-286.4, 24.7]$
Participant-specific κ_i minus benchmark, updating	249.3	$[106.9, 412.7]$
Plan-guided minus average endpoint, by adjective family		
Size–colour	202.4	$[169.3, 237.2]$
responses without form	29.6	$[9.1, 53.0]$
responses with form	172.8	$[138.9, 209.1]$
Size–form	150.7	$[108.5, 193.7]$
Colour–form	63.2	$[9.0, 115.6]$
Updating minus fixed, plan-guided, by colour reliability		
$\nu_{\mathrm{C}} = .59$	-61.7	$[-80.9, -43.6]$
$\nu_{\mathrm{C}} = .75$	-45.9	$[-59.9, -32.0]$
Without condition-specific length and selection terms		
Plan-guided minus global, averaged over regimes	286.1	$[90.5, 444.6]$
Plan-guided minus fully incremental, averaged over regimes	488.6	$[308.6, 710.3]$
Updating minus fixed, plan-guided	-71.4	$[-94.5, -49.7]$
Terminal utility instead of prefix utility		
Global, terminal minus prefix utility	-934.1	$[-1135.3, -749.9]$
Plan-guided, terminal minus prefix utility	58.3	$[-72.4, 200.0]$
Plan-guided minus global	1257.6	$[979.9, 1578.3]$
Updating minus fixed, plan-guided	-12.5	$[-17.6, -7.7]$
Without stable order term		
Plan-guided, without minus with the term	-3585.3	$[-3953.6, -3212.4]$
Updating minus fixed, plan-guided	23.5	$[11.7, 37.5]$
New participants (ten-fold cross-validation)		
Plan-guided minus global	271.6	$[79.5, 425.1]$
Plan-guided minus fully incremental	457.8	$[264.8, 694.3]$
Participant-specific κ_i minus shared κ	318.8	$[213.4, 455.2]$
Updating minus fixed, plan-guided	-53.8	$[-72.6, -34.8]$

D Participant Variation and Predictive Contributions

The joint model estimates pragmatic optimality, κ_i, and stable-order weight for each participant. Table 8 reports its predictive gain relative to the otherwise matched shared-κ model under each semantic regime. In the best-predicting context-fixed model, the population location of the participant-specific κ_i was $\mu_\kappa = .472$, with a 95% credible interval of $[.337, .618]$. The participant distribution was broad, with an estimated logit-scale standard deviation of 2.720, 95% credible interval $[2.277, 3.207]$.

Participants whose posterior mean κ_i lay farther from the shared $\kappa = .446$ gained more participant-level ELPD from the hierarchy. The observed correlation was $r = .618$, and the participant-level Bayesian-bootstrap mean was .639, with a 95% credible interval of $[.525, .763]$ and posterior probability above zero greater than .999. At the response level, the hierarchy gained 229.372 ELPD on two-adjective utterances, 95% credible interval $[163.012, 319.334]$, and 183.796 ELPD on three-adjective utterances, 95% credible interval $[100.757, 304.826]$. Its contribution to one-adjective utterances was -48.426 ELPD (95% credible interval $[-70.371, -28.666]$).

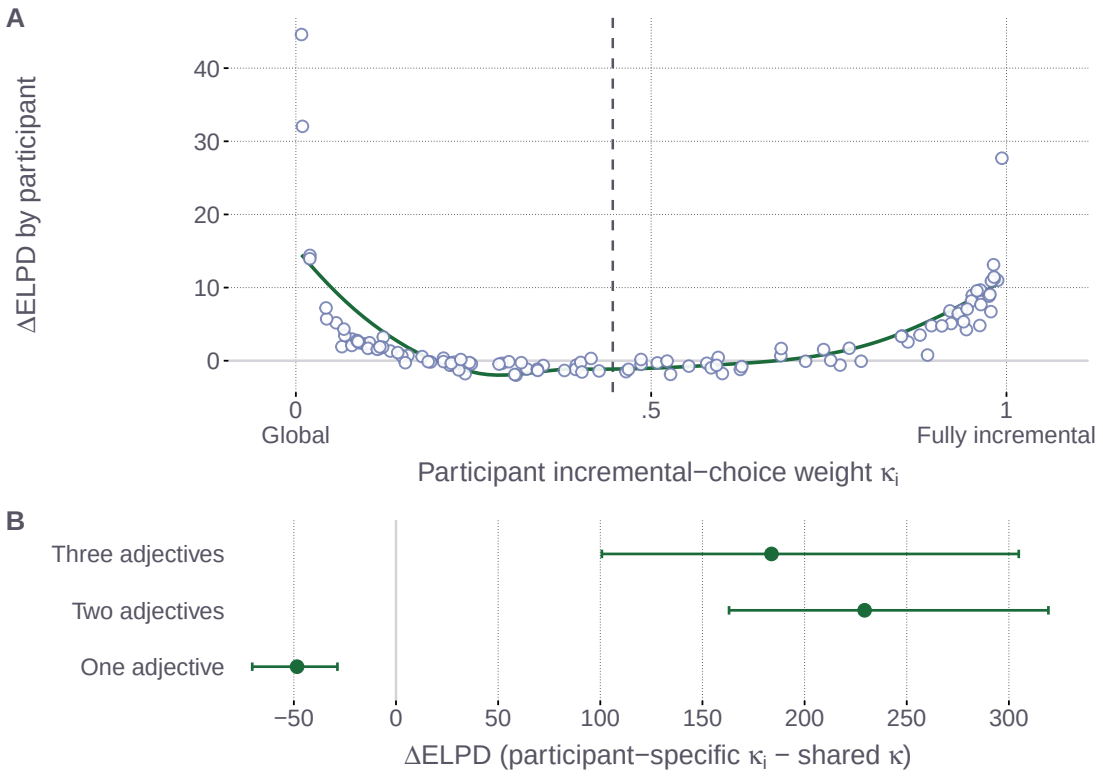


Figure 13: Where participant-specific κ_i improve prediction. Panel A relates each participant's leave-one-out gain from the κ_i hierarchy to their posterior mean κ_i; the dashed line marks the shared-κ estimate and the curve is a descriptive smooth. Panel B decomposes the total ELPD gain by observed response length. Intervals in Panel B are 95% credible intervals from the participant-level Bayesian bootstrap. Both panels use the best-predicting plan-guided model and its otherwise matched shared-κ model to localise the held-out model contrast.

E Local Normalisation and Model Endpoints

Local adjective competition. Substituting Equations (3) and (4) into Equation (5) and suppressing the shared referent and context arguments, the shared terms $\beta\Omega(u)+\Lambda(u,C)$ enter with weight one, and the remaining terms combine into the accumulated utility minus κ times the local-normalisation term:

$$\begin{aligned}(1-\kappa)\sum_{t=0}^{T-1} U(w_{t+1}\mid w_{1:t}) + \kappa\sum_{t=0}^{T-1}\log\frac{\exp U(w_{t+1}\mid w_{1:t})}{\sum_{w'\in\mathcal{A}(w_{1:t})}\exp U(w'\mid w_{1:t})}\\ = \sum_{t=0}^{T-1} U(w_{t+1}\mid w_{1:t}) - \kappa\sum_{t=0}^{T-1}\log\sum_{w'\in\mathcal{A}(w_{1:t})}\exp U(w'\mid w_{1:t}).\end{aligned} \tag{16}$$

The normalising constants of S_{G} and S_{I} do not depend on u and are absorbed by the renormalisation over the fifteen completed utterances.

Relation between the endpoints. At common parameter values, plan-guided production with $\kappa = 0$ exactly recovers global production, and with $\kappa = 1$ exactly recovers fully incremental production. The stable-order score, length and selection terms, and utterance space are identical across the three architectures. The interpolation in Equation (5) applies before the final Laplace smoothing described in Appendix B.

F Sequential Context Updating for Size

For the hierarchical semantic path, let L_{t+1} be the interpretation after the syntactically inner modifiers have composed. The semantic regime determines the comparison class used for size:

$$[\![\mathrm{S}]\!]_{\mathrm{fix}}(o) = [\![\mathrm{S}]\!]_{\theta_\delta(C)}(o), \tag{17a}$$

$$[\![\mathrm{S}]\!]_{\mathrm{upd},L_{t+1}}(o) = [\![\mathrm{S}]\!]_{\theta_\delta(L_{t+1})}(o). \tag{17b}$$

where θ_δ derives a graded size threshold from a comparison class and C is the original display, represented by the uniform prior P_0. Under updating, the literal interpretation recomputes the size threshold from the hierarchically prior state; under context-fixed semantics, it anchors the threshold to the display prior. At each surface position, the production calculation evaluates the current prefix using this noun-outward interpretation.

G Simulation Parameter Sweeps

The original two-order simulation compares a global rule based on completed-utterance utility with an incremental rule that includes continuation-versus-stop competition. We report its context construction, definitions, and parameter sweeps first. Appendix G.1 then introduces the production specification on the same contexts, as in the controlled comparison in Section 5.

Each random display contains between 2 and 30 objects whose sizes are drawn from a normal distribution centred at 15.5 and truncated to $[1, 30]$. The largest object is designated as the target and assigned blue; each non-target object is independently assigned blue or non-blue with equal probability. Three levels of contextual size spread vary how distinctly the largest target differs from its competitors. The semantic grid varies colour reliability, size-threshold selectivity, and perceptual blur, which control the strength of colour-based restriction, how selectively *big* applies, and how sharply size differences are represented. In the simulation notation, k corresponds to threshold selectivity δ and w_f to perceptual blur w in Equations (7) and (8).

Figures 15 to 17 show how the three fixed semantic parameters affect communicative success across the two speaker architectures. Each figure separates the semantic regimes and averages equally over the other parameter settings. For the two-order simulation, write $q(v, o) = L(o \mid v, C)^\alpha$, with $\alpha = 1$ and equal utterance costs. For distinct adjectives a, b, denote the original global and incremental rules by S_G and S_I:

$$S_G(ab \mid o, C) = \frac{q(ab, o)}{q(ab, o) + q(ba, o)}, \tag{18}$$

$$R_I(ab \mid o, C) = \frac{q(a, o)}{q(a, o) + q(b, o)} \frac{q(ab, o)}{q(ab, o) + q(a, o)}, \tag{19}$$

$$S_I(ab \mid o, C) = \frac{R_I(ab \mid o, C)}{R_I(ab \mid o, C) + R_I(ba \mid o, C)}. \tag{20}$$

The second factor of R_I is the continuation-versus-stop choice, and the final normalisation conditions on producing a completed two-adjective order. The production global model accumulates utility across adjective positions; its incremental endpoint assigns utterance length through the completed-response distribution. The pragmatic listener uses a uniform referent prior, giving $L^{\mathrm{prag}}(o \mid ab, C) = S(ab \mid o, C) / \sum_r S(ab \mid r, C)$.

The reproducible sweep uses seed 20260906 and samples 1,000 independent contexts for each combination of the number of objects and size spread. Each context is reused across semantic parameters, semantic regimes, and speakers, yielding 24,000 independent contexts and 12,000,000 model evaluations. Monte Carlo standard errors are calculated after averaging parameter settings within contexts; repeated evaluations of one context do not count as independent samples.

Figure 14 shows how the size-first preference changes across the broader random contexts, in which colour generally restricts the comparison class without uniquely identifying the target. Wider size distributions generally make the largest target more distinguishable from its competitors; clustered sizes reduce this distinction.

The original global rule with context-fixed semantics provides a symmetry baseline: both orders have the same fixed intersective meaning and equal cost, so they yield the same target probability. Sequential context updating breaks this symmetry because *big blue* interprets size relative to the comparison class established by colour, producing a size-first preference under global production. The preference is strongest in small contexts with widely separated sizes, where restricting the comparison class has the clearest consequences for which objects count as big, and weakens as contexts grow or sizes cluster.

The original incremental rule adds a contribution from local informativity along the surface sequence. With closely clustered sizes, this rule yields a colour-first preference in pragmatic target recovery. As sizes spread apart,

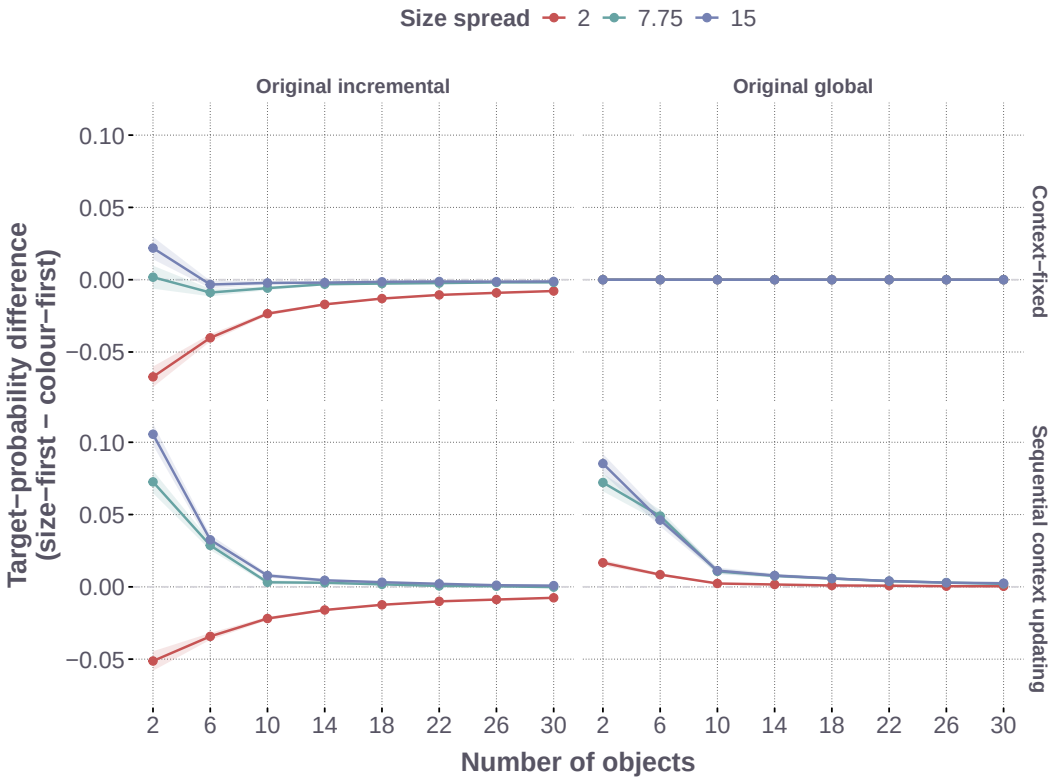


Figure 14: Size-first preference Δ_{SF} across context size and contextual spread, shown for the original two-order global and incremental rules under context-fixed semantics and sequential context updating. Wider size distributions make size a more reliable identifying cue; clustered distributions make it less reliable. Each point averages over the semantic sensitivity settings; ribbons show ± 1.96 Monte Carlo standard errors calculated after averaging those settings within each sampled context. The complete parameter grid appears in Table 12. Positive values indicate a higher model-assigned target probability for the size-first order.

the colour-first preference weakens. After averaging over the semantic parameter settings in Figure 14, the original incremental rule yields colour-first preferences in some contexts; the original global rule remains symmetric under fixed semantics and yields a size-first preference under updating. Under the original incremental rule, sequential context updating increases the size-first preference in target recovery when the colour-restricted comparison class changes size interpretation sufficiently strongly. The parameter sweeps in Figures 15 to 17 show that the same competition extends across colour reliability, size-threshold selectivity, and perceptual blur.

Table 12: Simulation parameter grid

Parameter	**Grid values**	**# levels**
Original two-order rule	Incremental, global	2
Semantic regime	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
Colour reliability ν	0.90, 0.92, 0.94, 0.96, 0.98	5
Threshold selectivity k	0.10, 0.30, 0.50, 0.70, 0.90	5
Perceptual blur w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Contexts per object-number/spread pair	1,000	
Total model evaluations	**12,000,000**	

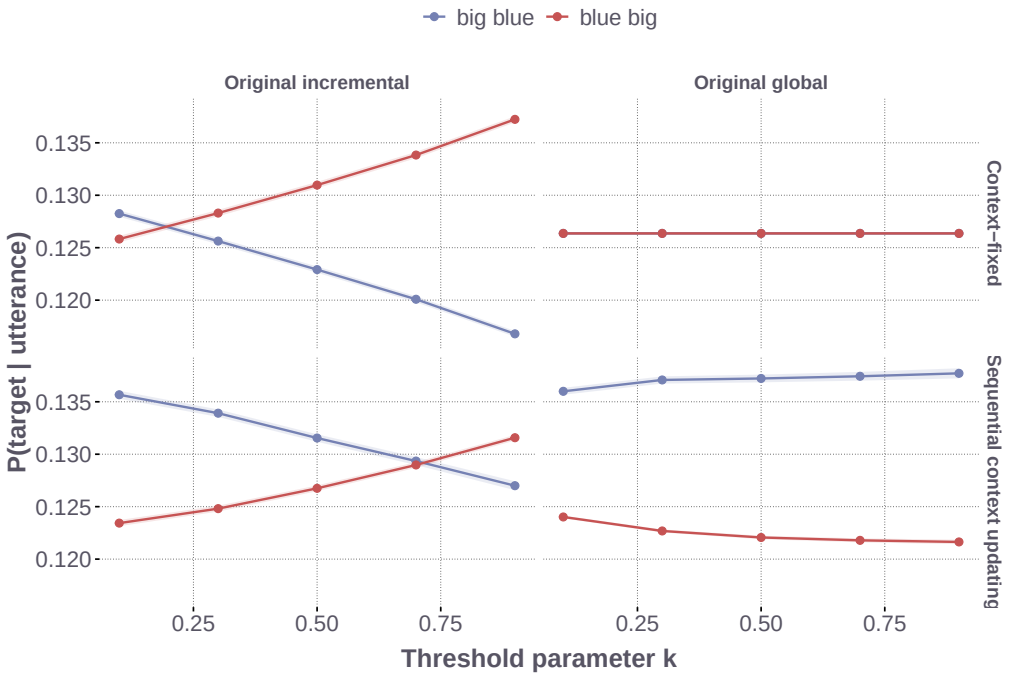


Figure 15: Effect of the threshold parameter k on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 context-level Monte Carlo standard errors.

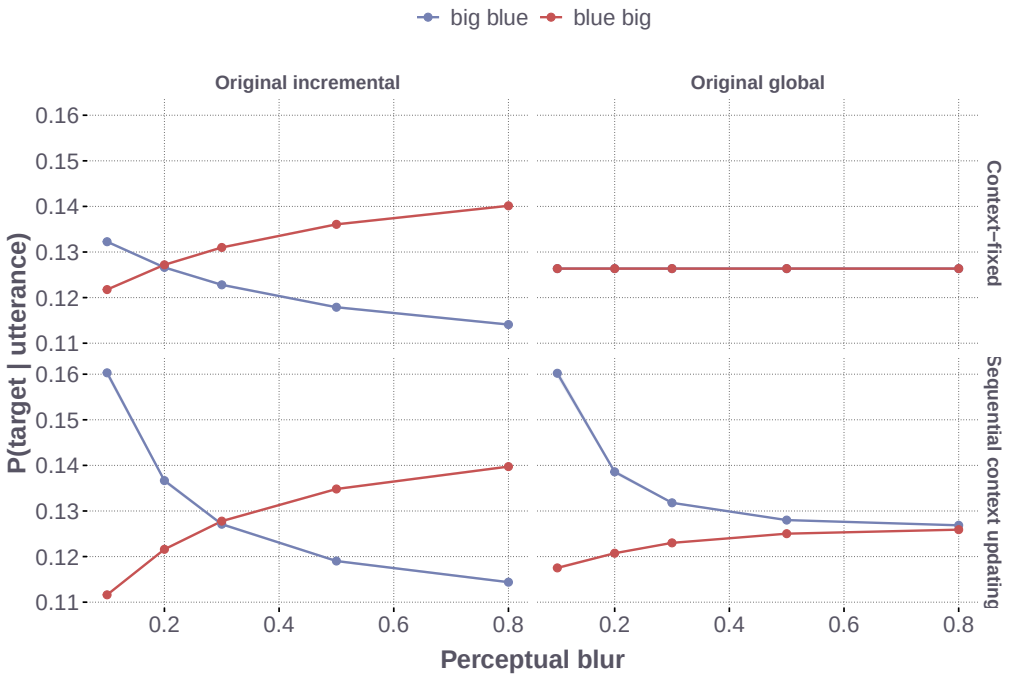


Figure 16: Effect of perceptual blur w_f on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 context-level Monte Carlo standard errors.

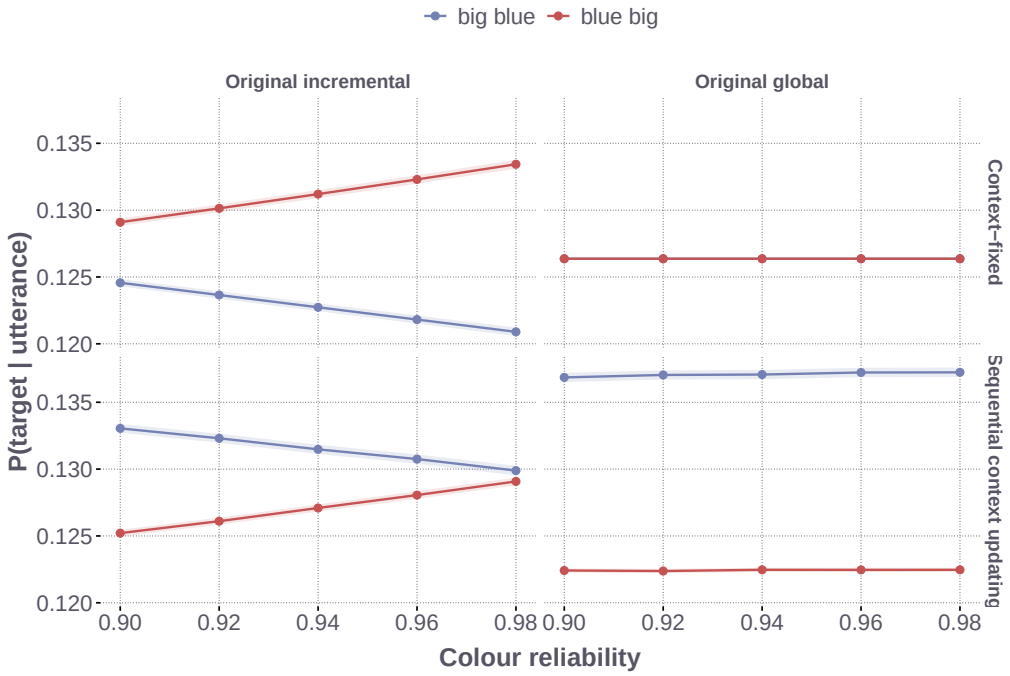


Figure 17: Effect of colour reliability ν on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 context-level Monte Carlo standard errors.

G.1 Controlled Changes on the Same Random Contexts

The terminal utility differs from the prefix utility:

$$U_{\text{term}}(u, o, C) = \alpha \log L(o \mid u, C), \tag{21}$$

$$U_{\text{prefix}}(u, o, C) = \alpha \sum_{t=1}^{|u|} \log L(o \mid w_{1:t}, C). \tag{22}$$

Both scores are globally normalised over completed utterances. The prefix score with $\kappa = 0$ is the global model used in the main text; $\kappa = .5$ and $\kappa = 1$ give the plan-guided and fully incremental simulation models. The terminal score supplies a control for the additional contribution of intermediate adjective information.

The controlled comparisons reuse all 24,000 contexts from the original sweep and the target-recovery measure in Equation (6). Each adjective retains the same extension when the intended referent changes; the referent prior is uniform. The response spaces contain two orders, four size–colour responses including the single adjectives, or the fifteen production responses. For expanded spaces, the pragmatic listener uses each order's probability under the full speaker distribution, without first conditioning on the two orders. The comparisons include the terminal utility and the prefix utility at $\kappa = 0, .5, 1$, using the original 125 semantic settings or the production constants. Prefix scores and local candidate sets are recomputed for each utterance space. They retain the production convention of no separate STOP action and final normalisation over completed paths. This isolates the computational components without assigning fitted participant effects or experimental task labels to random contexts.

The main comparison holds pragmatic optimality at $\alpha = 1$ and excludes stable-order expectations, salience, length and selection terms, and final Laplace smoothing. Form reliability is fixed at .50. Figure 7 shows the six speaker models under the three sets of alternatives and the semantic parameter grid, with the global speaker in the two-order condition implemented as in the earlier simulations. The production constants are colour reliability .59, threshold selectivity .50, and perceptual blur .6856. Table 13 reports the mean preferences in both panels of Figure 7. The prefix-utility rows specify the architecture through κ; the terminal utility uses global production. For the fifteen-response global model at the production constants, the paired updating-minus-fixed difference was 0.0037 percentage points, with a Monte Carlo interval of $[0.0007, 0.0067]$.

Table 13: Mean size-first preference in pragmatic target recovery in percentage points on the same 24,000 random contexts.

Utility	Responses	Semantic settings	Context-fixed	Updating
Terminal	2	Parameter grid	0.000	1.141
Prefix ($\kappa = 0$)	2	Parameter grid	-2.718	-1.838
Prefix ($\kappa = 0$)	4	Parameter grid	2.257	2.772
Prefix ($\kappa = 0$)	15	Parameter grid	2.589	3.104
Prefix ($\kappa = 0$)	15	Production constants	1.891	1.894
Prefix ($\kappa = .5$)	15	Production constants	1.455	1.457
Prefix ($\kappa = 1$)	15	Production constants	0.960	0.962

The main figure uses the production semantic implementation, whose probability floors differ slightly from those of the original sweep in Figure 14.

Figure 18 reports the fifteen-response comparison across utility rules and local-normalisation weights. The final panel adds the same stable-order coefficient, $\exp(.962357)$, to every model at the production semantic constants. This setting is the fixed model's population log-weight location. At $\kappa = 0$, adding this term gives mean preferences of 2.154 and 2.157 percentage points for fixed and updating semantics, respectively. The positive preferences are already present without this term. All other salience, length, and selection terms remain zero, and $\alpha = 1$ throughout.

For every contrast, semantic parameter settings are averaged within each context before calculating Monte Carlo uncertainty. The overall mean gives equal weight to the 24 context-size and size-spread groups.

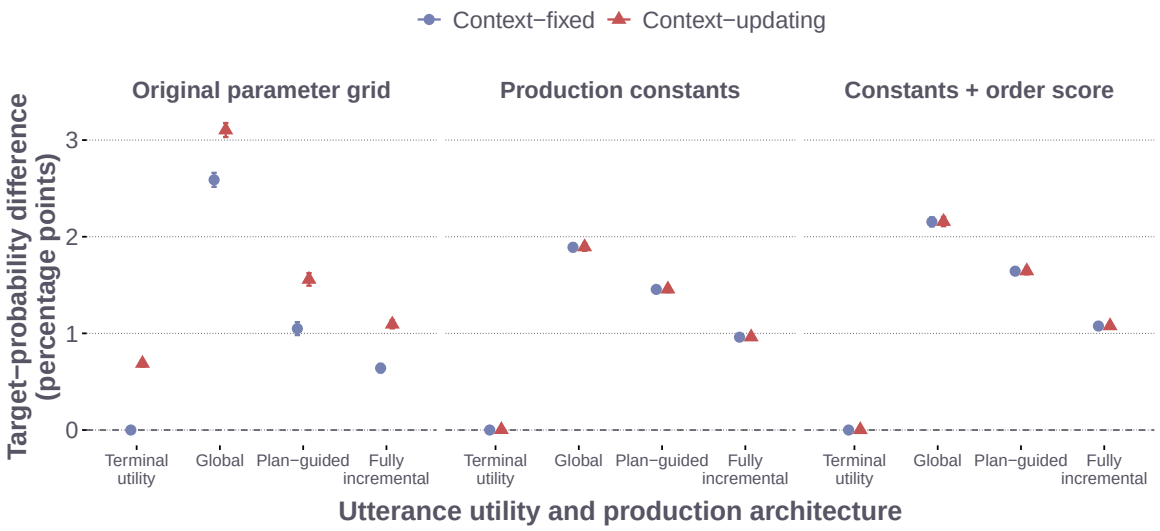


Figure 18: Controlled fifteen-response comparison across utility rules, semantic parameters, and a shared stable-order score. The terminal utility uses only the final literal target probability. The production models accumulate prefix scores: $\kappa = 0$ gives global production, $\kappa = .5$ plan-guided production, and $\kappa = 1$ fully incremental production. Panels use the original semantic grid, the production semantic constants, and those constants with the stable-order score added. Points and intervals show mean target-probability differences and ± 1.96 context-level Monte Carlo standard errors. All comparisons use the same random contexts.

G.2 Postnominal Variant of the Controlled Simulation

The postnominal variant reuses the contexts, semantic settings, and utility rules of Table 13. Only the direction of composition changes: the noun-adjacent adjective is the first adjective of the utterance, so sequential context updating proceeds in surface order. The prefix utility follows the surface order in both versions. Context-fixed meanings do not depend on the direction of composition, so the context-fixed column applies to both versions.

Table 14: Mean size-first preference in pragmatic target recovery in percentage points for prenominal and postnominal composition on the same 24,000 random contexts.

Utility	Responses	Semantic settings	Context-fixed	Updating, prenominal	Updating, postnominal
Terminal	2	Parameter grid	0.000	1.141	−1.141
Prefix ($\kappa = 0$)	2	Parameter grid	−2.718	−1.838	−3.603
Prefix ($\kappa = 0$)	4	Parameter grid	2.257	2.772	1.830
Prefix ($\kappa = 0$)	15	Parameter grid	2.589	3.104	2.156
Prefix ($\kappa = 0$)	15	Production constants	1.891	1.894	1.887
Prefix ($\kappa = .5$)	15	Production constants	1.455	1.457	1.452
Prefix ($\kappa = 1$)	15	Production constants	0.960	0.962	0.959

Under the terminal utility, the postnominal preference is the mirror image of the prenominal one. In the fifteen-response space, updating lowers the size-first preference in the postnominal version and raises it in the prenominal version by similar amounts. At the production semantic constants, updating changes the preference by less than 0.01 percentage points in either version. Monte Carlo 95% intervals are at most ± 0.17 percentage points. The contexts are regenerated from the seed of the original sweep by `simulation/run_postnominal_variant.py`.

G.3 Supplementary Predictions on Experimental Displays

The supplementary simulation uses the size–colour trials and all eight fitted production models. For each model, 50 evenly spaced retained draws from each of its four chains give 200 posterior parameter draws. The simulation preserves the fifteen named responses, fitted semantic constants, participant parameters, stable-order score, salience calculation, and length and selection terms. For each trial and posterior draw, each of the six objects is evaluated as the intended referent while the adjective extensions and observed task covariates remain fixed. Thus changing the candidate referent does not change which colour or form the uttered words denote.

Let $S_{in}^{(d)}(u \mid o, C)$ be the resulting complete-response probability at posterior draw d. We evaluate the speaker's size-first probability conditional on the size–colour adjective set as

$$\frac{S_{in}^{(d)}(\mathrm{SC} \mid o^*, C)}{S_{in}^{(d)}(\mathrm{SC} \mid o^*, C) + S_{in}^{(d)}(\mathrm{CS} \mid o^*, C)}. \tag{23}$$

Separately, the pragmatic listener normalises the full-response speaker probabilities over the six referents using a uniform prior. The listener therefore retains the fitted fifteen-response normalisation when evaluating each order. We average each quantity over the original size–colour trials within posterior draw, then report the posterior mean and central 95% interval.

Figure 19 reports the resulting speaker and listener predictions. In the plan-guided models with participant-specific κ_i, the mean size-first preference in target probability was $.0062$ under fixed semantics, 95% posterior interval $[.0047, .0077]$, and $.0058$ under updating, $[.0045, .0072]$. The corresponding mean probabilities of size-first conditional on the size–colour adjective set were $.967$ and $.966$. Both fitted regimes therefore generate a size-first preference under this counterfactual listener calculation. In the response-score decomposition, the predictive difference between these models was concentrated in adjective content and length. The two calculations describe how the models differ in predicting which utterance is produced while both assigning a listener preference for size-first order.

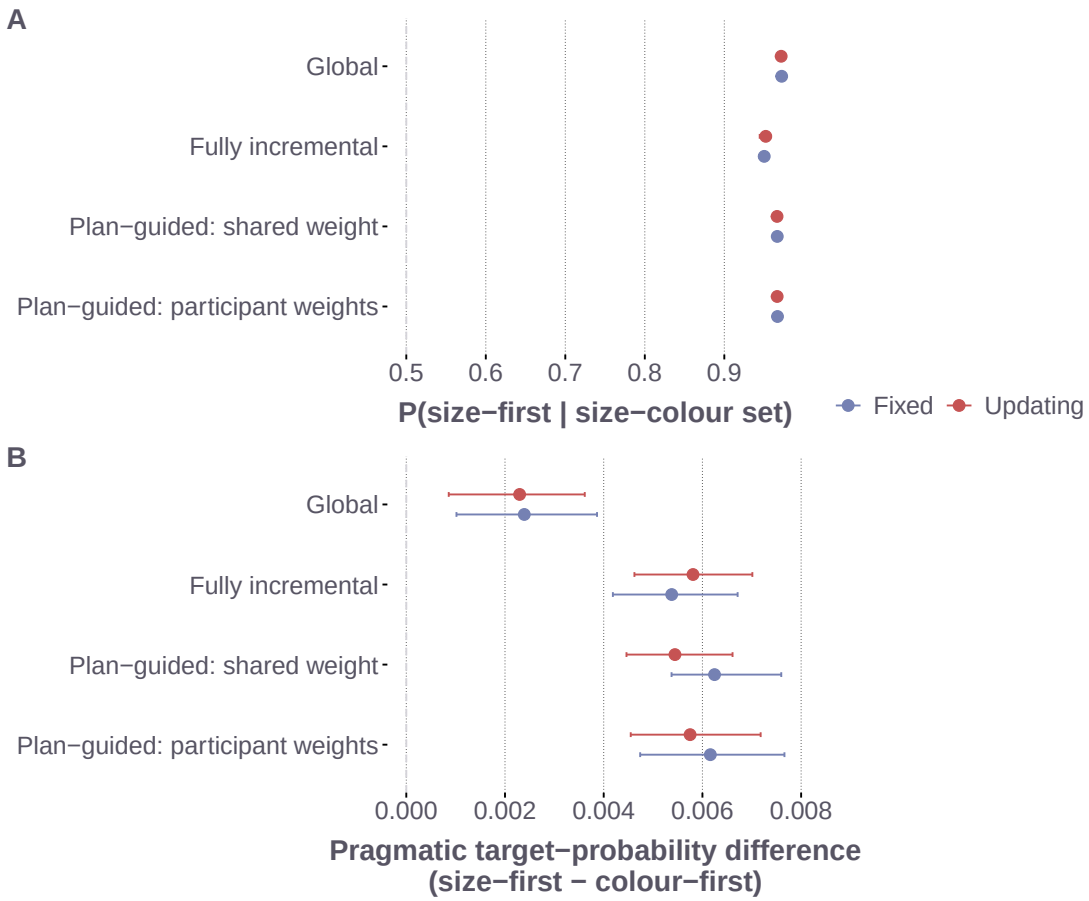


Figure 19: Supplementary predictions from the fitted production models on the size–colour displays. Panel A shows mean speaker probability of size-first conditional on selecting the size–colour adjective set. Panel B shows the difference in pragmatic target probability between size-first and colour-first. Both panels preserve the fitted fifteen-response distribution; the listener in Panel B uses a uniform prior over six referents. Intervals are central 95% posterior intervals over trial-averaged quantities using 200 posterior parameter draws per model.